

# Discrete-Time Fourier Series

## 4.1 INTRODUCTION

In the previous chapter, we studied the ways of representing a continuous-time periodic signal as a sum of sinusoids or exponentials. In this chapter, we will focus on discrete-time signals. While the discussion closely resembles the description used for continuous-time signals, there are some important differences. In particular, the Fourier series representation of a discrete-time periodic signal is a finite series, as opposed to the infinite series representation required for continuous-time periodic signals. As a consequence, there are no mathematical issues of convergence.

## 4.2 DISCRETE-TIME FOURIER SERIES (DTFS)

A discrete-time signal  $x(n)$  is periodic with period  $N$  if

$$x(n) = x(n + N) \quad (4.1)$$

for some positive integer  $N$ . The smallest value of  $N$  for which this equation holds is the *fundamental period*. The fundamental frequency is  $\omega_0 = 2\pi/N$  rad/sample. A periodic signal  $x(n)$  can be represented by a discrete-time Fourier series made up of sinusoids (or exponentials) of fundamental frequency  $\omega_0 = 2\pi/N$  and its harmonics. Because of its compactness and ease of mathematical manipulations, the exponential form is preferable to the trigonometric. The complex exponential  $e^{j(2\pi/N)n}$  is periodic with period  $N$ . The exponential Fourier series consists of the exponentials  $e^{j0n}$ ,  $e^{\pm j\omega_0 n}$ ,  $e^{\pm j2\omega_0 n}$ , ...,  $e^{jk\omega_0 n}$  ... and so on. The set of all discrete-time complex exponential signals that are periodic with period  $N$  is given by

$$\phi_k(n) = e^{jk\omega_0 n} = e^{jk(2\pi/N)n}, \quad k = 0, \pm 1, \pm 2, \dots \quad (4.2)$$

All these signals have fundamental frequencies that are multiples of  $2\pi/N$  and thus are harmonically related. The discrete-time periodic signal  $x(n)$  can be represented by a summation of complex exponentials  $\phi_k(n)$  of the form

$$x(n) = \sum_k X_k \phi_k(n) = \sum_k X_k e^{jk\omega_0 n} = \sum_k X_k e^{jk(2\pi/N)n} \quad (4.3)$$

As mentioned in Section 1.6.2 [Eq. (1.41)], the discrete-time exponentials whose frequencies are separated by  $2\pi$  (or integer multiples of  $2\pi$ ) are identical.

Specifically,

$$\begin{aligned}\phi_0(n) &= e^{j0(2\pi/N)n} = \phi_N(n) = e^{jN(2\pi/N)n} \\ \phi_1(n) &= e^{j(2\pi/N)n} = \phi_{N+1}(n) = e^{j(N+1)(2\pi/N)n}\end{aligned}$$

and, in general

$$\phi_k(n) = \phi_{k+N}(n) \quad (4.4)$$

Thus, the first harmonic is identical to the  $(N+1)$ st harmonic, the second harmonic is identical to the  $(N+2)$ nd harmonic, and so on. In other words, there are only  $N$  independent harmonics, and their frequencies range over an interval  $2\pi$  (because the harmonics are separated by  $\omega_0 = 2\pi/N$ ). This means, that the discrete-time Fourier series has only a finite number  $N$  of terms, unlike their continuous-time counterpart. This means, we may choose  $N$  independent harmonics  $e^{jk\omega_0 n}$  over  $0 \leq k \leq (N-1)$ , or over  $-1 \leq k \leq (N-2)$ , or over  $1 \leq k \leq N$ , or over any other suitable choice for that matter. Everyone of these sets will have the same harmonics, although in different order. Therefore, the Fourier series of a periodic signal  $x(n)$  consists of only  $N$  harmonics and can be expressed as

$$x(n) = \sum_{k=k_0}^{k_0+N-1} X_k e^{jk\omega_0 n} \quad (4.5)$$

where  $k_0$  is arbitrary. Since  $k_0$  is arbitrary, we can use the notation

$$x(n) = \sum_{k=\langle N \rangle} X_k e^{jk\omega_0 n} \quad (4.6)$$

where  $\omega_0 = 2\pi/N$ , and  $\sum_{k=\langle N \rangle}$  means summation over any range of consecutive  $k$ 's exactly

$N$  in length. Equation (4.6) is referred to as the *discrete-time Fourier series* and the coefficients  $X_k$  as the *Fourier series coefficients*.

## 4.2.1 Evaluation of DTFS Coefficients

To determine the Fourier series coefficients  $X_k$ , we replace the summation variable  $k$  by  $m$  on the right side of Eq. (4.6) and multiply both sides by  $e^{-jk\omega_0 n}$  to get

$$x(n)e^{-j\omega_0 kn} = \sum_{m=\langle N \rangle} X_m e^{j\omega_0(m-k)n}$$

Then, we sum over values of  $n$  in  $[0, N-1]$  to get

$$\sum_{n=0}^{N-1} x(n)e^{-j\omega_0 kn} = \sum_{n=0}^{N-1} \sum_{m=\langle N \rangle} X_m e^{j\omega_0(m-k)n} \quad (4.7)$$

By interchanging the order of summation in Eq. (4.7), we can write

$$\sum_{n=0}^{N-1} x(n)e^{-j\omega_0 kn} = \sum_{m=\langle N \rangle} X_m \sum_{n=0}^{N-1} e^{j\omega_0(m-k)n} \quad (4.8)$$

We know that

$$\sum_{n=0}^{N-1} \alpha^n = \frac{1 - \alpha^N}{1 - \alpha}, \quad \alpha \neq 1 \quad (4.9)$$

For  $\alpha = 1$ , we have

$$\sum_{n=0}^{N-1} \alpha^n = N \quad (4.10)$$

If  $m - k$  is not an integer multiple of  $N$  (i.e.,  $(m - k) \neq rN$  for  $r = 0, \pm 1, \pm 2, \dots$ ), we can let  $\alpha = e^{j\omega_0(m-k)n}$  in Eq. (4.9) to get

$$\sum_{n=0}^{N-1} e^{j\omega_0(m-k)n} = \frac{1 - e^{j\omega_0(m-k)N}}{1 - e^{j\omega_0(m-k)}} = \frac{1 - e^{j2\pi/N(m-k)N}}{1 - e^{j2\pi/N(m-k)}} = 0 \quad (4.11)$$

If  $(m - k)$  is an integer multiple of  $N$ , we can use Eq. (4.10), so that we have

$$\sum_{n=0}^{N-1} e^{j\omega_0(m-k)n} = N \quad (4.12)$$

Combining Eqs (4.11) and (4.12), we write

$$\sum_{n=0}^{N-1} e^{j\omega_0(m-k)n} = N\delta(m - k - rN) \quad (4.13)$$

where  $\delta(m - k - rN)$  is the unit sample occurring at  $m = k + rN$ . Substitution into Eq. (4.8) then yields

$$\sum_{n=0}^{N-1} x(n)e^{-j\omega_0 kn} = \sum_{m=\langle N \rangle} X_m N\delta(m - k - rN) \quad (4.14)$$

Since the summation on the right side is carried out over  $N$  consecutive values of  $m$  for a fixed value of  $k$ , it is clear that the only value that  $r$  can take in the given range of summation is  $r = 0$ . Thus, the only nonzero value in the sum corresponds to  $m = k$ , and the right-hand side of Eq. (4.14) evaluates to  $NX_k$ , so that

$$X_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n)e^{-j\omega_0 kn} \quad (4.15)$$

Because each of the terms in the summation in Eq. (4.15) is periodic with period  $N$ , the summation can be taken over any  $N$  successive values of  $n$ . We thus have the pair of equations

$$x(n) = \sum_{k=\langle N \rangle} X_k e^{jk\omega_0 n} \quad (4.16)$$

and

$$X_k = \frac{1}{N} \sum_{n=\langle N \rangle} x(n)e^{-jk\omega_0 n} \quad (4.17)$$

which together form the discrete-time Fourier series pair. Equation (4.16) is referred to as the *synthesis* equation and Eq. (4.17) as the *analysis* equation. As in continuous-time periodic signal, the discrete-time Fourier series coefficients  $X_k$  are often referred to as the *spectral coefficients* of  $x(n)$ . These coefficients specify a decomposition of  $x(n)$  into a sum of  $N$  harmonically related complex exponentials.

Because the Fourier series for discrete-time periodic signals is a finite sum defined entirely by the values of the signal over one period, the series always converges. The Fourier series provides an exact alternative representation of the time signal, and issues such as convergence or the Gibbs phenomenon do not arise.

### 4.2.2 Magnitude and Phase Spectrum of Discrete-Time Periodic Signals (Fourier Spectra)

The Fourier series consists of  $N$  components

$$X_0, X_1 e^{j\omega_0 n}, X_2 e^{j2\omega_0 n}, \dots, X_{N-1} e^{j(N-1)\omega_0 n}$$

The frequencies of these components are  $0, \omega_0, 2\omega_0, \dots, (N-1)\omega_0$  where  $\omega_0 = 2\pi/N$ . The amount of the  $k$ th harmonic is  $X_k$  (the Fourier coefficient). We can plot this amount  $X_k$  as a function of index  $k$  or  $\omega$ . Such a plot is called the *Fourier spectrum* of  $x(n)$ .

In general, the Fourier coefficients  $X_k$ , are complex, and they can be represented in the polar form as

$$X_k = |X_k| e^{j\angle X_k} \quad (4.18)$$

The plot of  $|X_k|$  versus  $\omega$  is called the magnitude spectrum and that of  $\angle X_k$  versus  $\omega$  is called the angle (or phase) spectrum. These two plots together are the frequency spectra of  $x(n)$ . Knowing these spectra, we can reconstruct or synthesize  $x(n)$  according to Eq. (4.16).

The results are very similar to the representation of a continuous-time periodic signal by an exponential Fourier series except that, the bandwidth of the continuous-time periodic signal is infinite and consists of an infinite number of exponential components (harmonics). The spectrum of the discrete-time periodic signal, in contrast, is bandlimited and has at most  $N$  components.

The DTFS coefficients  $X_k$  are periodic with period  $N$ , i.e.,

$$X_{k+N} = X_k \quad (4.19)$$

**Proof** From Eq. (4.17), we have

$$X_k = \frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-jk\omega_0 n}$$

$$\begin{aligned}
X_{k+N} &= \frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-j(k+N)\omega_0 n} \\
&= \frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-jk\omega_0 n} e^{j(2\pi/N)Nn} \\
&= \frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-jk\omega_0 n} e^{j2\pi n} \\
&= \frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-jk\omega_0 n}
\end{aligned}$$

$$X_{k+N} = X_k$$

Along the  $k$  scale,  $X_k$  repeats at intervals of  $N$ , and along the  $\omega$  scale,  $X_k$  repeats every  $2\pi$  intervals.

**Example 4.1** Determine the Fourier series coefficients of the signal  $x(n)$  and plot its magnitude and phase spectrum.

$$x(n) = 1 + \sin\left(\frac{2\pi}{N}n\right) + 3\cos\left(\frac{2\pi}{N}n\right) + \cos\left(\frac{4\pi}{N}n + \frac{\pi}{2}\right)$$

### Solution

The given signal is periodic with period  $N$  and frequency  $\omega_0 = 2\pi/N$  and it can be written as

$$x(n) = 1 + \sin(\omega_0 n) + 3\cos(\omega_0 n) + \cos\left(2\omega_0 n + \frac{\pi}{2}\right)$$

$$\begin{aligned}
x(n) &= 1 + \left(\frac{e^{j\omega_0 n} - e^{-j\omega_0 n}}{2j}\right) + 3\left(\frac{e^{j\omega_0 n} + e^{-j\omega_0 n}}{2}\right) + \left(\frac{e^{j2\omega_0 n + \pi/2} + e^{-j2\omega_0 n + \pi/2}}{2}\right) \\
&= 1 + \left(\frac{3}{2} + \frac{1}{2j}\right)e^{j\omega_0 n} + \left(\frac{3}{2} - \frac{1}{2j}\right)e^{-j\omega_0 n} + \left(\frac{1}{2}e^{j\pi/2}\right)e^{j2\omega_0 n} + \left(\frac{1}{2}e^{-j\pi/2}\right)e^{-j2\omega_0 n}
\end{aligned}$$

Comparing the above equation with

$$x(n) = X_0 + X_1 e^{j\omega_0 n} + X_{-1} e^{-j\omega_0 n} + X_2 e^{j2\omega_0 n} + X_{-2} e^{-j2\omega_0 n}$$

we get

$$\begin{aligned}
X_0 &= 1 \\
X_1 &= \frac{3}{2} + \frac{1}{2j} = \frac{3}{2} - \frac{1}{2}j \\
X_{-1} &= \frac{3}{2} - \frac{1}{2j} = \frac{3}{2} + \frac{1}{2}j \\
X_2 &= \frac{1}{2}j \\
X_{-2} &= -\frac{1}{2}j
\end{aligned}$$

All other Fourier series coefficients in the interval of summation in the synthesis equation are zero. The magnitude of the Fourier coefficients are

$$|X_0| = 1, \quad |X_1| = |X_{-1}| = \frac{\sqrt{10}}{2}, \quad |X_2| = |X_{-2}| = \frac{1}{2}$$

The phase of the Fourier coefficients are

$$\angle X_0 = 0, \quad \angle X_1 = -\tan^{-1}(1/3), \quad \angle X_{-1} = \tan^{-1}(1/3)$$

$$\angle X_2 = \frac{\pi}{2}, \quad \angle X_{-2} = -\frac{\pi}{2}$$

The magnitude spectrum and the phase spectrum are depicted in Figs 4.1(a) and (b), respectively.

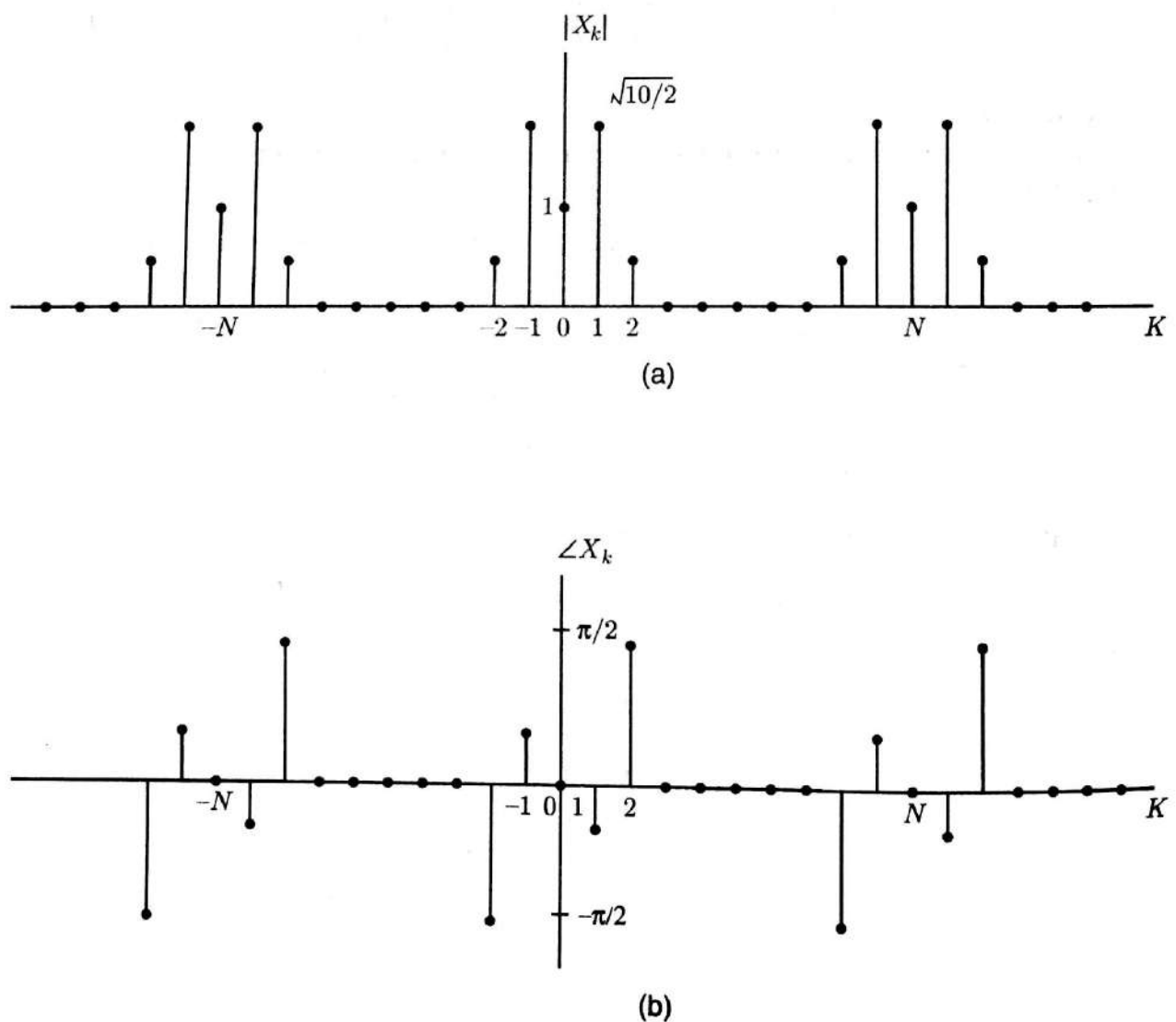


Fig. 4.1 (a) Magnitude spectrum, (b) Phase spectrum

**Example 4.2** Evaluate the Fourier series for the discrete-time periodic square wave  $x(n)$  shown in Fig. 4.2.

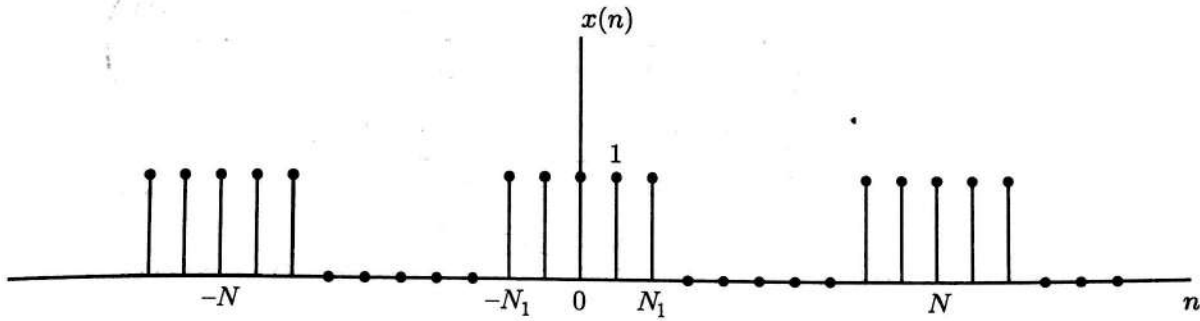


Fig. 4.2 Discrete-time periodic square wave

### Solution

The given signal is periodic with period  $N$  and frequency  $\omega_0 = 2\pi/N$  and it can be written as

$$x(n) = 1, \quad \text{for } -N_1 \leq n \leq N_1$$

It is convenient to choose the length- $N$  interval of summation in Eq. (4.17) so that it includes the range  $-N_1 \leq n \leq N_1$ . The Fourier coefficients are

$$X_k = \frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=-N_1}^{N_1} x(n) e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=-N_1}^{N_1} e^{-jk\omega_0 n}$$

A change of variables is performed by letting  $m = (n + N_1)$ , which also yields  $n = (m - N_1)$ ,  $m = 0$  as  $n = -N_1$ , and  $m = 2N_1$  as  $n = N_1$ . Therefore,

$$\begin{aligned} X_k &= \frac{1}{N} \sum_{m=0}^{2N_1} e^{-jk\omega_0(m-N_1)} \\ X_k &= \frac{1}{N} e^{jk\omega_0 N_1} \sum_{m=0}^{2N_1} e^{-jk\omega_0 m} \end{aligned} \quad (4.20)$$

Now, for  $k = 0, \pm N, \pm 2N, \dots$ , we have  $e^{jk\omega_0} = e^{-jk\omega_0} = 1$ , and Eq. (4.20) becomes

$$X_k = \frac{1}{N} \sum_{m=0}^{2N_1} 1 = \frac{2N_1 + 1}{N}, \quad k = 0, \pm N, \pm 2N, \dots$$

For  $k \neq 0, \pm N, \pm 2N, \dots$ , we may sum the geometric series in Eq. (4.20) to obtain

$$\begin{aligned}
 X_k &= \frac{1}{N} e^{jk\omega_0 N_1} \left( \frac{1 - e^{-jk\omega_0(2N_1+1)}}{1 - e^{-jk\omega_0}} \right) \\
 &= \frac{1}{N} e^{jk\omega_0 N_1} \left( \frac{e^{-jk\omega_0/2} (e^{jk\omega_0/2} - e^{-jk\omega_0(2N_1+1/2)})}{e^{-jk\omega_0/2} (e^{jk\omega_0/2} - e^{-jk\omega_0/2})} \right) \\
 &= \frac{1}{N} \frac{(e^{jk\omega_0(N_1+1/2)} - e^{-jk\omega_0(N_1+1/2)})}{(e^{jk\omega_0/2} - e^{-jk\omega_0/2})} \\
 X_k &= \frac{1}{N} \frac{\sin(k\omega_0(N_1 + 1/2))}{\sin(k\omega_0/2)}, \quad k \neq 0, \pm N, \pm 2N, \dots
 \end{aligned}$$

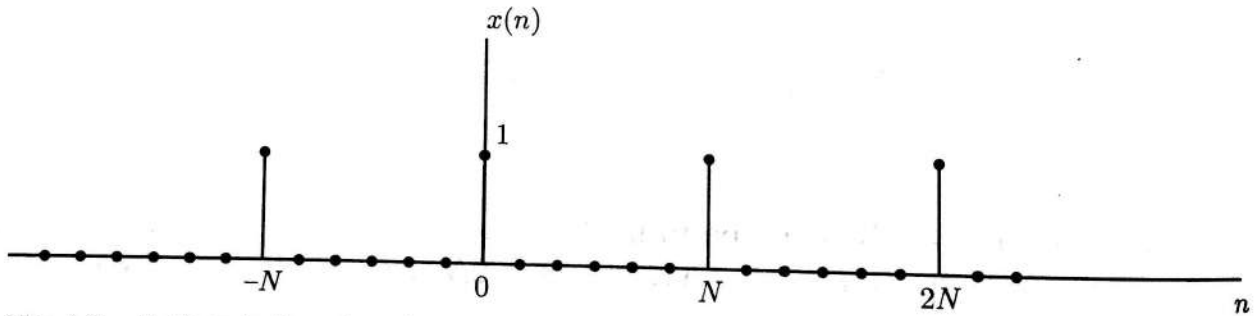
An alternative expression for  $X_k$  is obtained by substituting  $\omega_0 = 2\pi/N$ , yielding

$$X_k = \begin{cases} \frac{1}{N} \frac{\sin(k\pi(2N_1 + 1)/N)}{\sin(k\pi/N)}, & k \neq 0, \pm N, \pm 2N, \dots \\ \frac{2N_1 + 1}{N}, & k = 0, \pm N, \pm 2N, \dots \end{cases}$$

**Example 4.3** Find the DTFS coefficients of the  $N$ -periodic impulse train

$$x(n) = \sum_{m=-\infty}^{\infty} \delta(n - mN)$$

as shown in Fig. 4.3.



**Fig. 4.3** A discrete-time impulse train with time period  $N$

### Solution

The signal is periodic with period  $N$  and frequency  $\omega_0 = 2\pi/N$ . The given signal for one period may be written as

$$\begin{aligned}
 x(n) &= \begin{cases} 1, & n = 0 \\ 0, & 1 \leq n \leq N-1 \end{cases} \\
 &= \delta(n)
 \end{aligned}$$

Therefore, the Fourier series coefficients are

$$X_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=0}^{N-1} \delta(n) e^{-jk2\pi/N n} = \frac{1}{N} [e^{-jk(2\pi/N)n}]_{n=0} = \frac{1}{N}$$

**Example 4.4** Find the frequency domain representation of the signal depicted in Fig. 4.4.

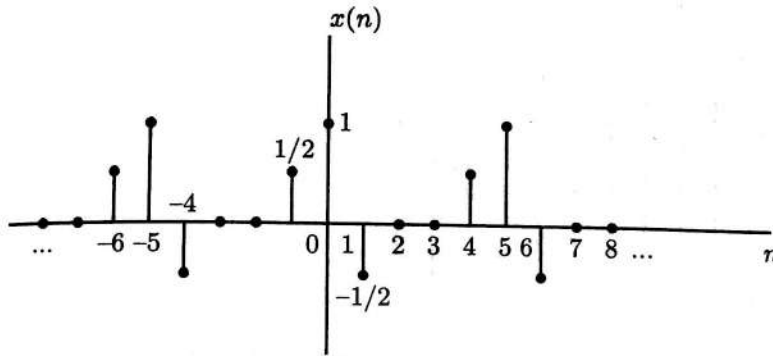


Fig. 4.4 Periodic signal

**Solution**

The signal has period  $N = 5$ , so  $\omega_0 = \frac{2\pi}{5}$ . Also, the signal has odd symmetry, so we sum over  $n = -2$  to  $n = 2$  in Eq. (4.17) to obtain  $X_k$ .

$$\begin{aligned} X_k &= \frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-jk\omega_0 n} \\ &= \frac{1}{5} \sum_{n=-2}^2 x(n) e^{-jk2\pi/5 n} \\ &= \frac{1}{5} \left[ x(-2) e^{jk4\pi/5} + x(-1) e^{jk2\pi/5} + x(0) + x(1) e^{-jk2\pi/5} + x(2) e^{-jk4\pi/5} \right] \end{aligned}$$

Using the values of  $x(n)$ , we get

$$X_k = \frac{1}{5} \left[ 1 + \frac{1}{2} e^{jk2\pi/5} - \frac{1}{2} e^{-jk2\pi/5} \right] = \frac{1}{5} \left[ 1 + j \sin\left(k \frac{2\pi}{5}\right) \right]$$

The one period of the DTFS coefficients  $X_k$ ,  $k = -2$  to  $k = 2$  are

$$X_{-2} = \frac{1}{5} - j \frac{\sin(4\pi/5)}{5} = 0.232 e^{-j0.531}$$

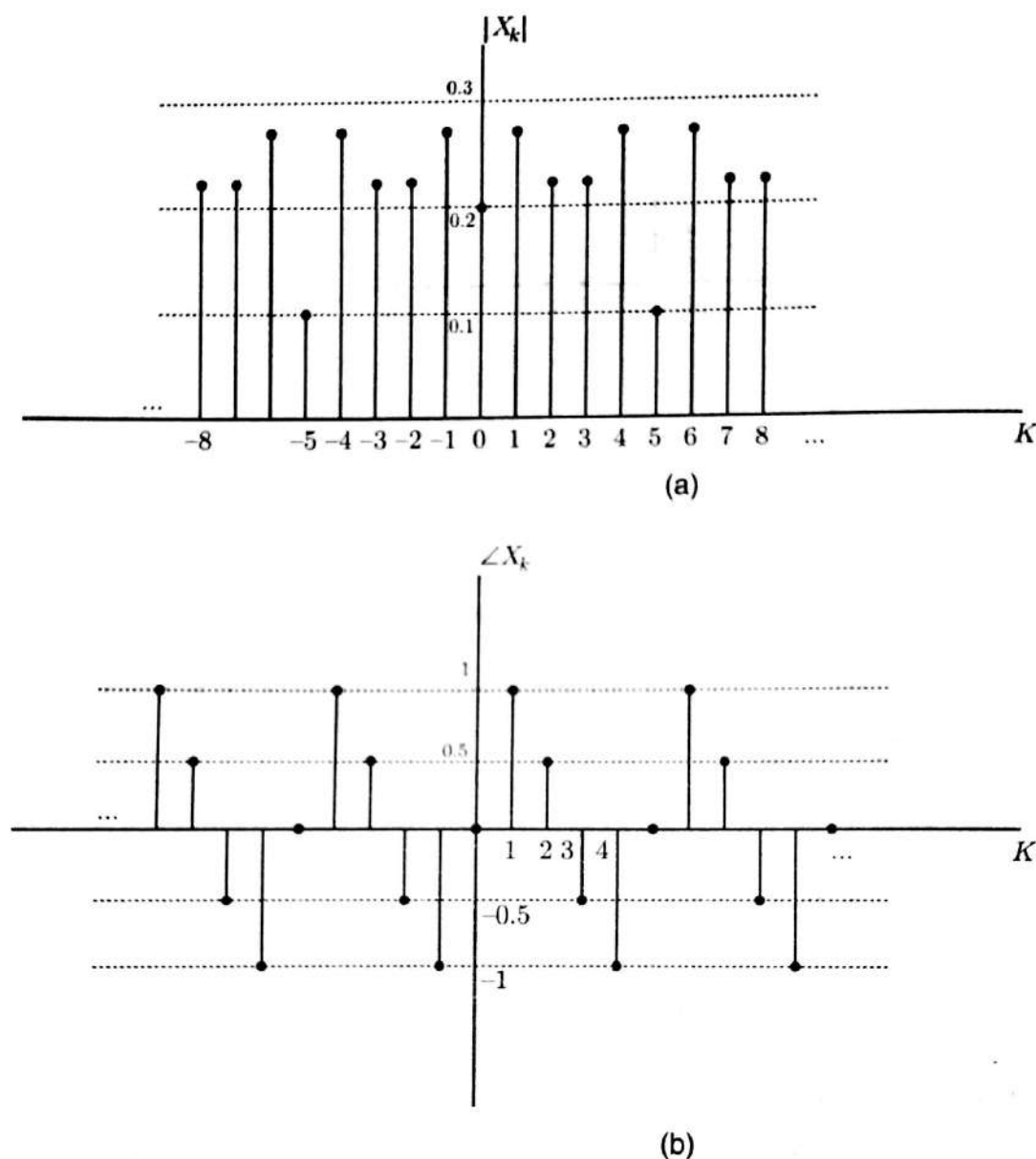
$$X_{-1} = \frac{1}{5} - j \frac{\sin(2\pi/5)}{5} = 0.276 e^{-j0.760}$$

$$X_0 = \frac{1}{5} = 0.2 e^{j0}$$

$$X_1 = \frac{1}{5} + j \frac{\sin(2\pi/5)}{5} = 0.276 e^{j0.760}$$

$$X_2 = \frac{1}{5} + j \frac{\sin(4\pi/5)}{5} = 0.232 e^{j0.531}$$

Figure 4.5 depicts the magnitude and phase of  $X_k$  as functions of frequency index  $k$ .



**Fig. 4.5** (a) Magnitude spectrum, (b) Phase spectrum

Now suppose we calculate  $X_k$  using  $n = 0$  to  $n = 4$  for the limits in Eq. (4.17)

$$X_k = \frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-jk\omega_0 n}$$

$$X_k = \frac{1}{5} \sum_{n=0}^4 x(n) e^{-jk(2\pi/5)n}$$

$$= \frac{1}{5} \left[ x(0) + x(1)e^{-jk(2\pi/5)} + x(2)e^{-jk(4\pi/5)} + x(3)e^{-jk(6\pi/5)} + x(4)e^{-jk(8\pi/5)} \right]$$

Using the values of  $x(n)$ , we get

$$X_k = \frac{1}{5} \left[ 1 - \frac{1}{2} e^{-jk(2\pi/5)} + \frac{1}{2} e^{-jk(8\pi/5)} \right]$$

Note that

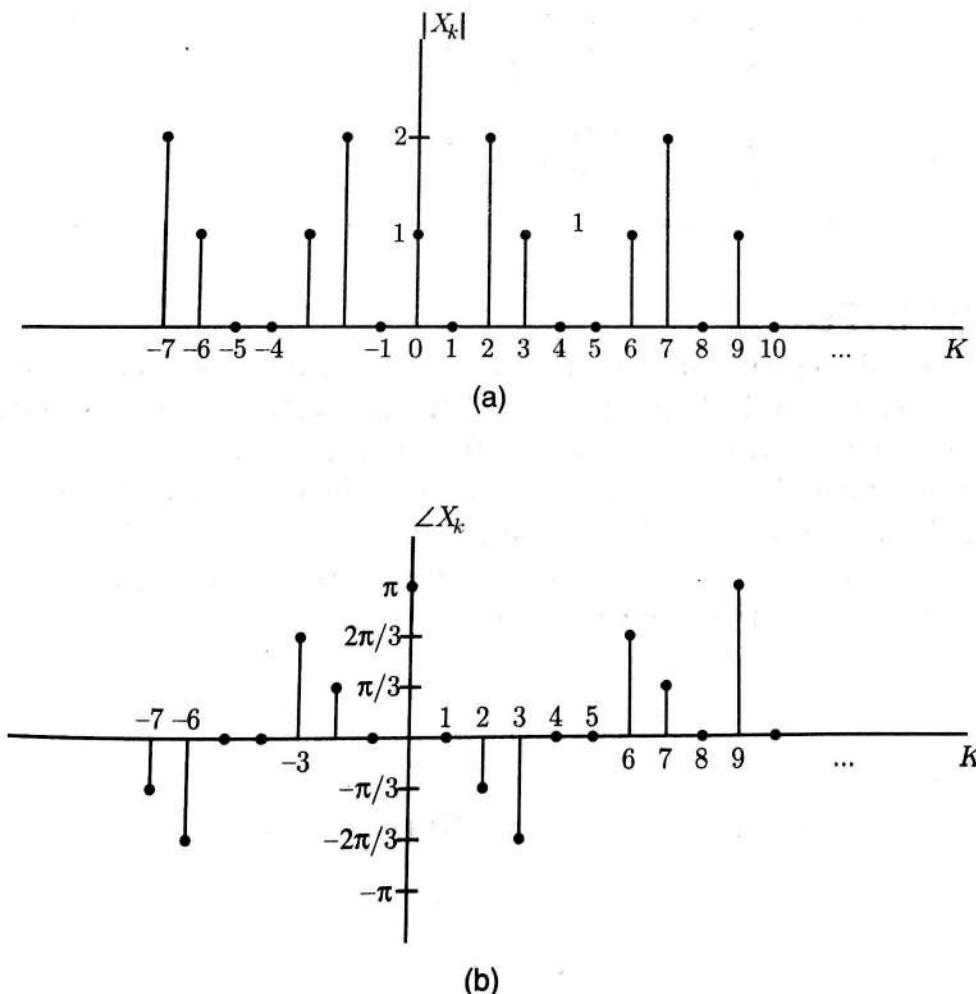
$$e^{-jk(8\pi/5)} = e^{-jk2\pi} e^{jk(2\pi/5)} = e^{jk(2\pi/5)}$$

Therefore,

$$\begin{aligned} X_k &= \frac{1}{5} \left[ 1 + \frac{1}{2} e^{jk(2\pi/5)} - \frac{1}{2} e^{-jk(2\pi/5)} \right] \\ &= \frac{1}{5} \left[ 1 + j \sin\left(k \frac{2\pi}{5}\right) \right] \end{aligned}$$

we see that both intervals,  $-2 \leq n \leq 2$  and  $0 \leq n \leq 4$  yield equivalent expressions for the DTFS coefficients.

**Example 4.5** Determine the time-domain signal  $x(n)$  from the DTFS coefficients depicted in Fig. 4.6.



**Fig. 4.6** (a) Magnitude spectrum, (b) Phase spectrum

**Solution**

The DTFS coefficients have period  $N = 9$ , so  $\omega_0 = 2\pi/9$ . From Fig. 4.6, the DTFS coefficients over the interval  $k = -4$  to  $k = 4$  are

$$\begin{aligned} X_k &= |X_k|e^{j\angle X_k}, \quad -4 \leq k \leq 4 \\ X_{-4} &= 0, \quad X_{-3} = 1e^{j2\pi/3}, \quad X_{-2} = 2e^{j\pi/3}, \quad X_{-1} = 0, \\ X_0 &= 1e^{j\pi}, \quad X_1 = 0, \quad X_2 = 2e^{-j\pi/3}, \quad X_3 = 1e^{-j2\pi/3}, \quad X_4 = 0 \end{aligned}$$

It is convenient to evaluate Eq. (4.16) over the interval  $k = -4$  to  $k = 4$  to obtain  $x(n)$ .

$$\begin{aligned} x(n) &= \sum_{k=\langle N \rangle} X_k e^{jk\omega_0 n} \\ &= \sum_{k=-4}^4 X_k e^{jk2\pi/9n} \\ &= X_{-4}e^{-j8\pi/9n} + X_{-3}e^{-j6\pi/9n} + X_{-2}e^{-j4\pi/9n} + X_{-1}e^{-j2\pi/9n} \\ &\quad + X_0 + X_1e^{j2\pi/9n} + X_2e^{j4\pi/9n} + X_3e^{j6\pi/9n} + X_4e^{j8\pi/9n} \\ &= e^{j2\pi/3}e^{-j6\pi/9n} + 2e^{j\pi/3}e^{-j4\pi/9n} + e^{j\pi} + 2e^{-j\pi/3}e^{j4\pi/9n} + e^{-j2\pi/3}e^{j6\pi/9n} \\ &= 2 \left( \frac{e^{j(6\pi/9n-2\pi/3)} + e^{-j(6\pi/9n-2\pi/3)}}{2} \right) + 4 \left( \frac{e^{j(4\pi/9n-\pi/3)} + e^{-j(4\pi/9n-\pi/3)}}{2} \right) - 1 \\ x(n) &= 2 \cos \left( \frac{6\pi}{9}n - \frac{2\pi}{3} \right) + 4 \cos \left( \frac{4\pi}{9}n - \frac{\pi}{3} \right) - 1 \end{aligned}$$

### 4.3 PROPERTIES OF DTFS

There are striking similarities between the properties of discrete-time and continuous-time Fourier series. The derivations of many of the properties are very similar to those of the corresponding properties of continuous-time Fourier series.

Similar to continuous-time series, it is often convenient to use a shorthand notation in DTFS to indicate the relationship between a periodic signal and its Fourier series coefficients. Specifically, if  $x(n)$  is a periodic signal with period  $N$  and with Fourier series coefficients denoted by  $X_k$ , then we will write

$$x(n) \longleftrightarrow X_k$$

#### 4.3.1 Linearity

If  $x(n)$  and  $y(n)$  denote two periodic signals with period  $N$ , and

$$x(n) \longleftrightarrow X_k$$

$$y(n) \longleftrightarrow Y_k$$

then

$$z(n) = ax(n) + by(n) \longleftrightarrow Z_k = aX_k + bY_k \quad (4.21)$$

**Proof** The Fourier series coefficients of  $z(n)$  is given by

$$\begin{aligned}
 Z_k &= \frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=\langle N \rangle} [ax(n) + by(n)] e^{-jk\omega_0 n} \\
 &= a \underbrace{\frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-jk\omega_0 n}}_{X_k} + b \underbrace{\frac{1}{N} \sum_{n=\langle N \rangle} y(n) e^{-jk\omega_0 n}}_{Y_k} \\
 &= aX_k + bY_k
 \end{aligned}$$

### 4.3.2 Time Shifting

When a time shift is applied to a periodic signal  $x(n)$ , the period  $N$  of the signal is preserved. If

$$x(n) \longleftrightarrow X_k$$

then  $y(n) = x(n - n_0) \longleftrightarrow Y_k = X_k e^{-jk\omega_0 n_0}$  (4.22)

One consequence of this property is that, when a signal is shifted in time, the *magnitudes* of its Fourier series coefficients remain unaltered. That is,  $|Y_k| = |X_k|$ .

**Proof** By definition,

$$Y_k = \frac{1}{N} \sum_{n=\langle N \rangle} y(n) e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=0}^{N-1} y(n) e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=0}^{N-1} x(n - n_0) e^{-jk\omega_0 n}$$

A change of variables is performed by letting  $m = (n - n_0)$ , which also yields  $(m \rightarrow -n_0)$  as  $(n \rightarrow 0)$ , and  $(m \rightarrow (N - 1 - n_0))$  as  $(n \rightarrow (N - 1))$ . Therefore,

$$\begin{aligned}
 Y_k &= \frac{1}{N} \sum_{m=-n_0}^{N-1-n_0} x(m) e^{-jk\omega_0(m+n_0)} \\
 Y_k &= \frac{1}{N} \sum_{m=-n_0}^{N-1-n_0} x(m) e^{-jk\omega_0 m} e^{-jk\omega_0 n_0} \\
 Y_k &= X_k e^{-jk\omega_0 n_0} \\
 |Y_k| &= |X_k|
 \end{aligned}$$

### 4.3.3 Frequency Shifting

If  $x(n) \longleftrightarrow X_k$

then  $y(n) = e^{jM\omega_0 n} x(n) \longleftrightarrow Y_k = X_{k-M}$  (4.23)

**Proof** By definition,

$$\begin{aligned}
 Y_k &= \frac{1}{N} \sum_{n=\langle N \rangle} y(n) e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=0}^{N-1} y(n) e^{-jk\omega_0 n} \\
 &= \frac{1}{N} \sum_{n=0}^{N-1} e^{jM\omega_0 n} x(n) e^{-jk\omega_0 n} \\
 &= \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j(k-M)\omega_0 n} = X_{k-M}
 \end{aligned}$$

Hence, a frequency shift corresponds to multiplication in time domain by a complex sinusoid whose frequency is equal to the time shift. ■

#### 4.3.4 Time Reversal

If  $x(n) \longleftrightarrow X_k$

then  $y(n) = x(-n) \longleftrightarrow Y_k = X_{-k}$  (4.24)

In other words, the time reversal applied to a discrete-time signal results in a time reversal of the corresponding sequence of Fourier series coefficients.

**Proof**

$$\begin{aligned}
 Y_k &= \frac{1}{N} \sum_{n=\langle N \rangle} y(n) e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=0}^{N-1} y(n) e^{-jk\omega_0 n} \\
 &= \frac{1}{N} \sum_{n=0}^{N-1} x(-n) e^{-jk\omega_0 n} = \frac{1}{N} \sum_{m=-(N-1)}^0 x(m) e^{-j(-k)\omega_0 m} = X_{-k}
 \end{aligned}$$

An interesting consequence of the time-reversal property is that if  $x(n)$  is even then its Fourier series coefficients are also even, i.e., if

$$x(-n) = x(n)$$

then  $X_{-k} = X_k$  (4.25)

Similarly, if  $x(n)$  is odd, then so are its Fourier series coefficients, i.e., if

$$x(-n) = -x(n)$$

then  $X_{-k} = -X_k$  (4.26) ■

#### 4.3.5 Time Scaling

Because of the discrete-time nature of the time index for discrete-time signals, the relation between time scaling and Fourier series coefficients in discrete-time takes on a different form from its continuous-time counterpart. If we try to define the signal  $x(an)$ , we run

into difficulties if  $a$  is not an integer. Therefore, we cannot slow down the signal by choosing  $a < 1$ . On the other hand, if we let  $a$  be an integer other than  $\pm 1$ , we do not merely speed up the original signal. For example, since  $n$  can take on only integer values, the signal  $x(2n)$  consists of the even samples of  $x(n)$  alone.

Let  $m$  be a positive integer, and the signal is defined as

$$x_{(m)}(n) = \begin{cases} x(n/m), & \text{if } n \text{ is a multiple of } m \\ 0, & \text{if } n \text{ is not a multiple of } m \end{cases} \quad (4.27)$$

$x_m(n)$  can be obtained from  $x(n)$  by placing  $(m-1)$  zeros between successive values of the original signal. Intuitively, we can think of  $x_{(m)}(n)$  as a slowed-down version of  $x(n)$ . If  $x(n)$  is periodic with period  $N$ , then  $y(n) = x_{(m)}(n)$  is also periodic with period  $mN$ . Now, if

$$x(n) \longleftrightarrow X_k$$

then

$$y(n) = x_{(m)}(n) \longleftrightarrow Y_k = \frac{1}{m} X_k \quad (4.28)$$

The Fourier series coefficients  $Y_k = (1/m) X_k$  are also periodic with period  $mN$ .

**Proof** The Fourier series coefficients of  $y(n) = x_{(m)}(n)$  [ $y(n)$  is periodic with period  $mN$  and fundamental frequency  $\omega_0/m$ ] are given by

$$\begin{aligned} Y_k &= \frac{1}{mN} \sum_{n=0}^{m(N-1)} y(n) e^{-jk(\omega_0/m)n} \\ &= \frac{1}{mN} \sum_{n=0}^{m(N-1)} x_{(m)}(n) e^{-jk(\omega_0/m)n} \\ &= \frac{1}{mN} \sum_{n=0}^{m(N-1)} x\left(\frac{n}{m}\right) e^{-jk(\omega_0/m)n} \end{aligned}$$

A change of variables is performed by letting  $r = n/m$ , which also yields  $r = 0$  as  $n = 0$ , and  $r = N-1$  as  $n = m(N-1)$ . Therefore,

$$Y_k = \frac{1}{m} \underbrace{\frac{1}{N} \sum_{r=0}^{N-1} x(r) e^{-jk\omega_0 r}}_{X_k} = \frac{1}{m} X_k$$

### 4.3.6 Periodic Convolution

If

$$x(n) \longleftrightarrow X_k,$$

$$y(n) \longleftrightarrow Y_k$$

then

$$z(n) = x(n) \circledast y(n) = \sum_{r=\langle N \rangle} x(r) y(n-r) \longleftrightarrow Z_k = N X_k Y_k \quad (4.29)$$

where  $\circledast$  denotes the periodic convolution.

**Proof** For periodic signals with the same period, a special form of convolution, known as periodic convolution, is defined as

$$z(n) = \sum_{r=\langle N \rangle} x(r)y(n-r)$$

It is easy to show that  $z(n)$  is periodic with period  $N$  and the periodic convolution is commutative and associative. Thus, we can write  $z(n)$  in a Fourier series representation with coefficients

$$Z_k = \frac{1}{N} \sum_{n=\langle N \rangle} z(n)e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=\langle N \rangle} \left( \sum_{r=\langle N \rangle} x(r)y(n-r) \right) e^{-jk\omega_0 n}$$

$$Z_k = \sum_{r=\langle N \rangle} x(r) \left( \frac{1}{N} \sum_{n=\langle N \rangle} y(n-r)e^{-jk\omega_0 n} \right)$$

From the time shifting property, i.e., if

$$y(n) \longleftrightarrow Y_k,$$

then

$$y(n-r) \longleftrightarrow Y_k e^{-jr\omega_0 n}$$

we have

$$Z_k = N \underbrace{\frac{1}{N} \sum_{r=\langle N \rangle} x(r)e^{-jr\omega_0 n}}_{X_k} Y_k = NX_k Y_k$$

The convolution in time transforms to multiplication of the frequency domain representations. ■

### 4.3.7 Multiplication

If  $x(n)$  and  $y(n)$  are periodic signals with the same period  $N$ , and

$$x(n) \longleftrightarrow X_k,$$

$$y(n) \longleftrightarrow Y_k,$$

then

$$z(n) = x(n)y(n) \longleftrightarrow Z_k = \sum_{r=\langle N \rangle} X_r Y_{k-r} \quad (4.30)$$

$Z_k$  may be interpreted as the periodic convolution between the two periodic sequences of Fourier coefficients  $X_k$  and  $Y_k$ .

**Proof** Consider the signal  $z(n)$ .

$$\begin{aligned} z(n) &= x(n)y(n) = \sum_{r=\langle N \rangle} X_r e^{jr\omega_0 n} \sum_{m=\langle N \rangle} Y_m e^{jm\omega_0 n} \\ &= \sum_{r=\langle N \rangle} X_r \sum_{m=\langle N \rangle} Y_m e^{j(m+r)\omega_0 n} \end{aligned}$$

A change of variables is performed by letting  $k = m + r$ , which also yields  $m = (k - r)$ ,  $(k \rightarrow r)$  as  $(m \rightarrow 0)$ , and  $[k \rightarrow (r + N - 1)]$  as  $[m \rightarrow (N - 1)]$ . Therefore,

$$\begin{aligned} z(n) &= \sum_{r=\langle N \rangle} X_r \sum_{k=r}^{r+N-1} Y_{k-r} e^{jk\omega_0 n} \\ &= \sum_{r=\langle N \rangle} X_r \sum_{k=\langle N \rangle} Y_{k-r} e^{jk\omega_0 n} \\ z(n) &= \sum_{k=\langle N \rangle} \left( \sum_{r=\langle N \rangle} X_r Y_{k-r} \right) e^{jk\omega_0 n} = \sum_{k=\langle N \rangle} Z_k e^{jk\omega_0 n} \end{aligned}$$

Thus,

$$Z_k = \sum_{r=\langle N \rangle} X_r Y_{k-r}$$

#### 4.3.8 First Difference

The discrete-time equivalent to the differentiation property of the continuous-time Fourier series involves the use of the first-difference operation, which is defined as  $y(n) = x(n) - x(n - 1)$ . If  $x(n)$  is periodic with period  $N$ , then so is  $y(n)$ , since shifting  $x(n)$  or linearly combining  $x(n)$  with another periodic signal whose period is  $N$  always results in a periodic signal with period  $N$ . Also, if

$$x(n) \longleftrightarrow X_k,$$

then

$$y(n) = x(n) - x(n - 1) \longleftrightarrow Y_k = (1 - e^{-jk\omega_0})X_k \quad (4.31)$$

A common use of this property is in situations where the evaluation of Fourier series coefficients is easier for the first difference than for the original sequence.

**Proof**

Given that

$$x(n) \longleftrightarrow X_k$$

Using the time-shifting property, we get

$$x(n - 1) \longleftrightarrow X_k e^{-jk\omega_0}$$

Now, using the linearity property, we get

$$x(n) - x(n - 1) \longleftrightarrow X_k - X_k e^{-jk\omega_0}$$

$$x(n) - x(n - 1) \longleftrightarrow (1 - e^{-jk\omega_0})X_k$$

### 4.3.9 Running Sum or Accumulation

If

$$x(n) \longleftrightarrow X_k$$

$$y(n) \longleftrightarrow Y_k$$

then

$$y(n) = \sum_{k=-\infty}^n x(k) \longleftrightarrow Y_k = \left( \frac{1}{1 - e^{-jk\omega_0}} \right) X_k, \quad k \neq 0 \quad (4.32)$$

The discrete-time Fourier series coefficient  $Y_k$  of the running sum  $y(n) = \sum_{k=-\infty}^n x(k)$  is finite-valued and periodic only if  $X_0 = 0$ .

**Proof** Consider the running sum

$$y(n) = \sum_{k=-\infty}^n x(k)$$

$$y(n) = x(n) + \sum_{k=-\infty}^{n-1} x(k)$$

$$y(n) = x(n) + y(n-1)$$

$$y(n) - y(n-1) = x(n)$$

Using the time-shifting and linearity properties, we get

$$Y_k - Y_k e^{-jk\omega_0} = X_k$$

$$Y_k = \left( \frac{1}{1 - e^{-jk\omega_0}} \right) X_k$$

### 4.3.10 Conjugation and Conjugate Symmetry

If

$$x(n) \longleftrightarrow X_k$$

then

$$y(n) = x^*(n) \longleftrightarrow Y_k = X_{-k}^* \quad (4.33)$$

Taking the complex conjugate of a periodic signal  $x(n)$  has the effect of complex conjugation and time reversal on the corresponding Fourier series coefficients.

**Proof**

$$\begin{aligned}
 Y_k &= \frac{1}{N} \sum_{n=\langle N \rangle} y(n) e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=\langle N \rangle} x^*(n) e^{-jk\omega_0 n} \\
 &= \left( \frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{jk\omega_0 n} \right)^* = \left( \underbrace{\frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-j(-k)\omega_0 n}}_{X_{-k}} \right)^* \\
 &= (X_{-k})^* = X_{-k}^*
 \end{aligned}$$

**Case I** If  $x(n)$  is real, i.e., if

$$x^*(n) = x(n)$$

then

$$X_{-k}^* = X_k$$

Therefore,

$$X_{-k} = X_k^* \quad (4.34)$$

In other words, the Fourier series coefficients will be *conjugate symmetric*. Also if  $x(n)$  is real and even, then from Eqs (4.25) and (4.34), we have

$$X_{-k} = X_k^* = X_k \quad (4.35)$$

That is, if  $x(t)$  is *real and even*, then so are its Fourier series coefficients.

**Case II** If  $x(n)$  is *real and odd*, then its Fourier series coefficients are purely imaginary and odd. Using Eqs (4.26) and (4.34), we get

$$X_{-k} = X_k^* = -X_k \quad (4.36)$$

**Case III** Even and odd decomposition of real signals: If  $x(n)$  is real and

$$x(n) \longleftrightarrow X_k$$

then

$$1. \quad x_e(n) = \mathcal{E}\{x(n)\} \longleftrightarrow \text{Re}\{X_k\} \quad (4.37)$$

The even part of a signal  $x(n)$  is defined as

$$x_e(n) = \frac{1}{2}[x(n) + x(-n)]$$

■

**Proof**

Using the linearity property, we get

$$\begin{aligned}x_e(n) &\longleftrightarrow \frac{1}{2}2\text{Re}\{X_k\} \\x_e(n) &\longleftrightarrow \text{Re}\{X_k\} \\x_e(n) &\longleftrightarrow \frac{1}{2}[X_k + X_{-k}] \\x_e(n) &\longleftrightarrow \frac{1}{2}[X_k + X_k^*]\end{aligned}$$

and

$$2. \quad x_o(n) = \mathcal{O}\{x(n)\} \longleftrightarrow j\text{Im}\{X_k\} \quad (4.38)$$

The odd part of a signal  $x(n)$  is defined as

$$x_o(n) = \frac{1}{2}[x(n) - x(-n)]$$

Using the linearity property, we get

$$\begin{aligned}x_o(n) &\longleftrightarrow \frac{1}{2}[X_k - X_{-k}] \\x_o(n) &\longleftrightarrow \frac{1}{2}[X_k - X_k^*] \\x_o(n) &\longleftrightarrow \frac{1}{2}2j\text{Im}\{X_k\} \\x_o(n) &\longleftrightarrow j\text{Im}\{X_k\}\end{aligned}$$

**4.3.11 Parseval's Relation**

If 
$$x(n) \longleftrightarrow X_k$$

then 
$$\frac{1}{N} \sum_{n=\langle N \rangle} |x(n)|^2 = \sum_{k=\langle N \rangle} |X_k|^2 \quad (4.39)$$

As in the continuous-time case, the left-hand side of Parseval's relation is the average power in one period for the periodic signal  $x(n)$ . Similarly,  $|X_k|^2$  is the average power in the  $k$ th harmonic component of  $x(n)$ . Thus, Parseval's relation states that the average power in a periodic signal equals the sum of the average powers in all of its harmonic components. In discrete-time, of course, there are only  $N$  distinct harmonic components, and since the  $X_k$  are periodic with period  $N$ , the sum on the right-hand side of Eq. (4.39) can be taken over any  $N$  consecutive values of  $k$ .

**Proof** Consider the LHS of Eq. (4.39), we have

$$\begin{aligned}
 \frac{1}{N} \sum_{n=\langle N \rangle} |x(n)|^2 &= \frac{1}{N} \sum_{n=\langle N \rangle} x(n) x^*(n) \\
 &= \frac{1}{N} \sum_{n=\langle N \rangle} x(n) \left( \sum_{k=\langle N \rangle} X_k e^{jk\omega_0 n} \right)^* \\
 &= \frac{1}{N} \sum_{n=\langle N \rangle} x(n) \left( \sum_{k=\langle N \rangle} X_k^* e^{-jk\omega_0 n} \right) \\
 &= \sum_{k=\langle N \rangle} X_k^* \left( \frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-jk\omega_0 n} \right) \\
 &= \sum_{k=\langle N \rangle} X_k^* X_k \\
 &= \sum_{k=\langle N \rangle} |X_k|^2
 \end{aligned}$$

#### 4.4 SYSTEMS WITH PERIODIC INPUTS

The response of an LTI system to a sinusoidal input leads to a characterization of system behaviour that is termed the *frequency response* of the system. This characterization is obtained in terms of the impulse response by using convolution and a complex sinusoidal input signal. Let the impulse response of a system be  $h(n)$  and the input be  $x(n) = e^{j\omega_0 n}$ . Then, the convolution integral gives the output as

$$\begin{aligned}
 y(n) &= h(n) * x(n) = \sum_{m=-\infty}^{\infty} h(m)x(n-m) = \sum_{m=-\infty}^{\infty} h(m)e^{j\omega_0(n-m)} \\
 &= \underbrace{\sum_{m=-\infty}^{\infty} h(m)e^{-j\omega_0 m}}_{H(e^{j\omega_0})} e^{j\omega_0 n} \\
 y(n) &= H(e^{j\omega_0})e^{j\omega_0 n} \tag{4.40}
 \end{aligned}$$

where we define

$$H(e^{j\omega_0}) = \sum_{m=-\infty}^{\infty} h(m)e^{-j\omega_0 m}$$

The output of the system is thus a complex sinusoid of the same frequency as the input multiplied by the complex number  $H(e^{j\omega_0})$  [Fig. 4.7(a)]. An intuitive interpretation of the sinusoidal steady-state response is obtained by writing the complex-valued frequency

response  $H(e^{j\omega_0})$  in polar form. We have

$$H(e^{j\omega_0}) = |H(e^{j\omega_0})|e^{j\angle H(e^{j\omega_0})} \quad (4.41)$$

where  $|H(e^{j\omega_0})|$  is now termed the *magnitude response* and  $\angle H(e^{j\omega_0})$  is termed the *phase response* of the system. Substituting this polar form into Eq. (4.39), we may express the output as

$$y(n) = |H(e^{j\omega_0})|e^{j(\omega_0 n + \angle H(e^{j\omega_0}))} \quad (4.42)$$

The system thus modifies the amplitude of the input by  $|H(e^{j\omega_0})|$  and the phase by  $\angle H(e^{j\omega_0})$ . Knowing  $H(e^{j\omega_0})$ , we can determine whether the system amplifies or attenuates a given sinusoidal component of the input and how much of a phase shift the system adds to that particular component.

We say that the complex sinusoid  $x(n) = e^{j\omega_0 n}$  is an *eigenfunction* of the system associated with the *eigenvalue*  $H(e^{j\omega_0})$ .

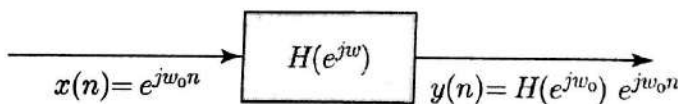
Signals that are eigenfunctions of systems play an important role in LTI system theory. By representing arbitrary signals as weighted superpositions of eigenfunctions, we transform the operation of convolution to multiplication. Thus, if the input to an LTI system is given by

$$x(n) = \sum_{k=\langle N \rangle} X_k e^{jk\omega_0 n} \quad (4.43)$$

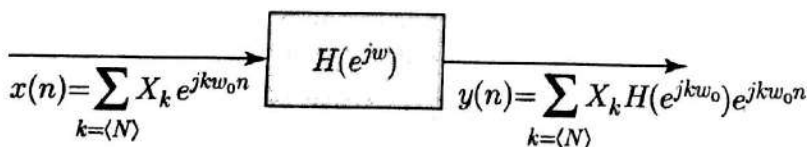
then, the output of the system is given by [Fig. 4.7(b)]

$$y(n) = \sum_{k=\langle N \rangle} X_k H(e^{jk\omega_0}) e^{jk\omega_0 n} = \sum_{k=\langle N \rangle} Y_k e^{jk\omega_0 n} \quad (4.44)$$

where  $\omega_0 = \frac{2\pi}{N}$  and  $Y_k = X_k H(e^{jk\omega_0})$ .



(a)



(b)

**Fig. 4.7** Response of an LTI system to complex sinusoid

# SOLVED EXAMPLES

**Example 4.6** A discrete-time periodic signal  $x(n]$  is real valued and has a fundamental period  $N = 5$ . The nonzero Fourier series coefficients for  $x(n]$  are

$$X_0 = 1, \quad X_2 = X_{-2}^* e^{j\pi/4}, \quad X_4 = X_{-4}^* = 2e^{j\pi/3}$$

Express  $x(n]$  in the form

$$x(n) = A_0 + \sum_{k=1}^{\infty} A_k \sin(\omega_k n + \phi_k)$$

## Solution

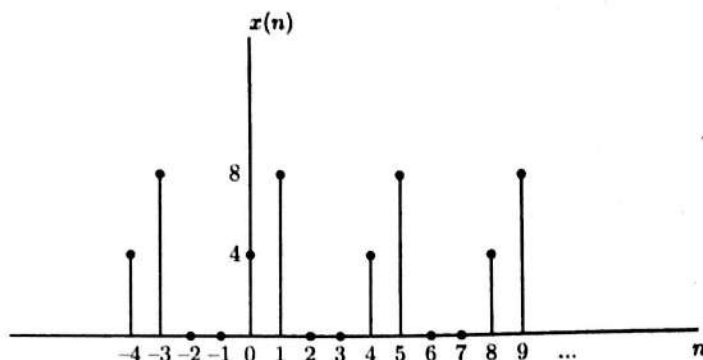
The signal  $x(n]$  is periodic with period  $N = 5$  and fundamental frequency  $\omega_0 = 2\pi/N = 2\pi/5$ . Using the Fourier synthesis Eq. (4.16), we have

$$\begin{aligned} x(n) &= \sum_{k=(N)} X_k e^{jk\omega_0 n} \\ &= X_0 + X_2 e^{j2\omega_0 n} + X_{-2} e^{-j2\omega_0 n} \\ &\quad + X_4 e^{j4\omega_0 n} + X_{-4} e^{-j4\omega_0 n} \\ &= X_0 + X_2 e^{j22\pi/Nn} + X_{-2} e^{-j22\pi/Nn} \\ &\quad + X_4 e^{j42\pi/Nn} + X_{-4} e^{-j42\pi/Nn} \\ &= 1 + e^{j\pi/4} e^{j22\pi/5n} + e^{-j\pi/4} e^{-j22\pi/5n} \\ &\quad + 2e^{j\pi/3} e^{j42\pi/5n} + 2e^{-j\pi/3} e^{-j42\pi/5n} \\ &= 1 + 2 \left( \frac{e^{j(4\pi/5n + \pi/4)} + e^{-j(4\pi/5n + \pi/4)}}{2} \right) \\ &\quad + 4 \left( \frac{e^{j(8\pi/5n + \pi/3)} + e^{-j(8\pi/5n + \pi/3)}}{2} \right) \\ &= 1 + 2\cos\left(\frac{4\pi}{5}n + \frac{\pi}{4}\right) + 4\cos\left(\frac{8\pi}{5}n + \frac{\pi}{3}\right) \\ &= 1 + 2\sin\left(\frac{4\pi}{5}n + \frac{3\pi}{4}\right) + 4\sin\left(\frac{8\pi}{5}n + \frac{5\pi}{6}\right) \\ &= A_0 + \sum_{k=1}^2 A_k \sin(\omega_k n + \phi_k) \end{aligned}$$

**Example 4.7** Evaluate the numerical values of one period of the Fourier series coefficients of the periodic signal

$$x(n) = \sum_{m=-\infty}^{\infty} [4\delta(n - 4m) + 8\delta(n - 1 - 4m)]$$

as shown in Fig. 4.8.



**Fig. 4.8** Periodic signal

## Solution

The signal is periodic with period  $N = 4$  and frequency  $\omega_0 = 2\pi/N = \pi/2$ . The given signal for one period may be written as

$$\begin{aligned} x(n) &= \begin{cases} 4, & n = 0 \\ 8, & n = 1 \\ 0, & n = 2, 3 \end{cases} \\ &= 4\delta(n) + 8\delta(n - 1) \end{aligned}$$

Therefore, the Fourier series coefficients are

$$\begin{aligned} X_k &= \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-jk\omega_0 n} \\ &= \frac{1}{4} \sum_{n=0}^3 x(n) e^{-jk\pi n/2} \\ &= \frac{1}{4} [4 + 8e^{-jk\pi/2}] \\ X_k &= 1 + 2e^{-jk\pi/2} \end{aligned}$$

The Fourier series coefficients  $X_k$  are also periodic ( $X_{k+N} = X_k$ ) with period  $N = 4$ . The numerical values of one period of  $X_k$ ,  $0 \leq k \leq 3$  are

$$\begin{aligned} X_0 &= 1 + 2 = 3 \\ X_1 &= 1 + 2e^{-j\pi/2} \\ &= 1 + 2(\cos(\pi/2) - j\sin(\pi/2)) = 1 - 2j \\ X_2 &= 1 + 2e^{-j\pi} \\ &= 1 + 2(\cos(\pi) - j\sin(\pi)) = 1 - 2 = -1 \\ X_3 &= 1 + 2e^{-j3\pi/2} \\ &= 1 + 2(\cos(3\pi/2) - j\sin(3\pi/2)) = 1 + 2j \end{aligned}$$

**Example 4.8** Let  $x(n)$  be a real and odd periodic signal with period  $N = 7$  and Fourier series coefficients  $X_k$ . Given that

$$X_{15} = j, \quad X_{16} = 2j, \quad X_{17} = 3j$$

determine the values of  $X_0, X_{-1}, X_{-2}, X_{-3}$ .

**Solution**

The discrete-time Fourier series coefficients are periodic with period  $N$ , i.e.,

$$X_k = X_{k+N} = X_{k+2N}$$

Therefore, for  $N = 7$ , we have

$$X_1 = X_8 = X_{15} = j$$

$$X_2 = X_9 = X_{16} = 2j$$

$$X_3 = X_{10} = X_{17} = 3j$$

Since the given signal  $x(n)$  is real and odd, the Fourier series coefficients  $X_k$  will be purely imaginary and odd ( $X_k = -X_{-k}$ ). Therefore,

$$X_0 = 0$$

$$X_1 = -X_{-1}$$

$$X_2 = -X_{-2}$$

$$X_3 = -X_{-3}$$

Finally, we have

$$X_{-1} = -X_1 = -j$$

$$X_{-2} = -X_2 = -2j$$

$$X_{-3} = -X_3 = -3j$$

**Example 4.9** Suppose we are given the following information about a signal  $x(n)$ :

1.  $x(n)$  is a real and even signal.
2.  $x(n)$  has a period  $N = 10$  and Fourier coefficients  $X_k$ .
3.  $X_{11} = 5$ .
4.  $\frac{1}{10} \sum_{n=0}^9 |x(n)|^2 = 50$ .

Show that  $x(n) = A \cos(Bn + C)$ , and specify numerical values for the constants  $A, B$ , and  $C$ .

**Solution**

The Fourier series coefficients are periodic with period  $N = 10$ , we have

$$X_k = X_{k+N}$$

$$X_k = X_{k+10}$$

$$X_1 = X_{11} = 5$$

$$X_{-1} = X_9$$

Since, the given signal  $x(n)$  is real and even, The Fourier coefficients  $X_k$  are also real and even. Therefore,

$$X_k = X_{-k}$$

$$X_1 = X_{-1} = X_9 = 5$$

Using Parseval's relation,

$$\frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2 = \sum_{k=0}^{N-1} |X_k|^2$$

$$\frac{1}{10} \sum_{n=0}^9 |x(n)|^2 = \sum_{k=0}^9 |X_k|^2 = 50$$

$$|x_0|^2 + |X_1|^2 + |X_9|^2 + \sum_{k=2}^8 |X_k|^2 = 50$$

$$|x_0|^2 + 25 + 25 + \sum_{k=2}^8 |X_k|^2 = 50$$

$$|X_0|^2 + \sum_{k=2}^8 |X_k|^2 = 0$$

Therefore, for one period the Fourier coefficients are

$$X_k = \begin{cases} 5, & k = 1, 9 \\ 0, & k = 0 \text{ and } 2 \leq k \leq 8 \end{cases}$$

Now using the synthesis Eq. (4.16), we have

$$x(n) = \sum_{k=\langle N \rangle} X_k e^{jk\omega_0 n}$$

$$= \sum_{k=0}^{N-1} X_k e^{jk2\pi/Nn} = \sum_{k=0}^9 X_k e^{jk2\pi/10n}$$

$$= X_1 e^{j2\pi/10n} + X_9 e^{j92\pi/10n}$$

Substituting  $X_9 = X_{-1}$ , we get

$$\begin{aligned} x(n) &= X_1 e^{j2\pi/10n} + X_{-1} e^{j-2\pi/10n} \\ &= 5e^{j\pi/5n} + 5e^{-j\pi/5n} \\ &= 10 \left( \frac{e^{j\pi/5n} + e^{-j\pi/5n}}{2} \right) \\ &= 10 \cos \left( \frac{n\pi}{5} \right) \end{aligned}$$

**Example 4.10** Each of the two sequences  $x(n)$  and  $y(n)$  has a period  $N = 4$ , and the corresponding Fourier series coefficients are specified as

$$x(n) \longleftrightarrow X_k, \quad y(n) \longleftrightarrow Y_k$$

where

$$\begin{aligned} X_0 &= X_3 = \frac{1}{2} X_1 = \frac{1}{2} X_2 = 1 \\ Y_0 &= Y_1 = Y_2 = Y_3 = 1 \end{aligned}$$

Using the multiplication property, determine the Fourier series coefficients  $Z_k$  for the signal  $z(n) = x(n)y(n)$ .

**Solution**

The given signal  $z(n) = x(n)y(n)$  is periodic with period  $N = 4$ . Using the multiplication property [Eq. (4.30)], we have

$$z(n) = x(n)y(n)$$

$$Z_k = \sum_{r=\langle N \rangle} X_r Y_{k-r}$$

$$Z_k = \sum_{r=0}^{N-1} X_r Y_{k-r}$$

$$Z_k = \sum_{r=0}^3 X_r Y_{k-r}$$

$$Z_k = X_0 Y_k + X_1 Y_{k-1} + X_2 Y_{k-2} + X_3 Y_{k-3}$$

$$Z_k = Y_k + 2Y_{k-1} + 2Y_{k-2} + Y_{k-3}$$

Since  $Y_k = Y_{k+N} = 1$  for all values of  $k$ , therefore

$$Z_k = 1 + 2 + 2 + 1 = 6 \quad \text{for all values of } k$$

**Example 4.11** When the impulse train

$$x(n) = \sum_{k=-\infty}^{\infty} \delta(n - 4k)$$

is the input to a particular LTI system with frequency response  $H(e^{j\omega})$ , the output of the system is found to be

$$y(n) = \cos \left( \frac{5\pi}{2} n + \frac{\pi}{4} \right)$$

Determine the values of  $H(e^{jk\pi/2})$  for  $k = 0, 1, 2$  and 3.

**Solution**

The given input signal  $x(n)$  is periodic with period  $N = 4$  and frequency  $\omega_0 = 2\pi/N = \pi/2$ . The given signal for one period may be written as

$$\begin{aligned} x(n) &= \begin{cases} 1, & n = 0 \\ 0, & n = 1, 2, 3 \end{cases} \\ &= \delta(n) \end{aligned}$$

Therefore, the Fourier series coefficients are

$$\begin{aligned} X_k &= \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-jk\omega_0 n} \\ &= \frac{1}{4} \sum_{n=0}^3 x(n) e^{-jk2\pi/4n} \\ &= \frac{1}{4} \sum_{n=0}^3 \delta(n) e^{-jk2\pi/4n} \\ X_k &= \frac{1}{4}, \quad \text{for all } k \end{aligned}$$

From Eq. (4.44), we know that the output  $y(n)$  of an LTI system to a periodic input is given by,

$$\begin{aligned} y(n) &= \sum_{k=\langle N \rangle} X_k H(e^{jk\omega_0}) e^{jk\omega_0 n} \\ &= \sum_{k=0}^{N-1} X_k H(e^{jk2\pi/N}) e^{jk2\pi/Nn} \\ &= \sum_{k=0}^3 X_k H(e^{jk2\pi/4}) e^{jk2\pi/4n} \\ y(n) &= X_0 H(e^{j0}) e^{j0n} + X_1 H(e^{j\pi/2}) e^{j\pi/2n} \\ &\quad + X_2 H(e^{j\pi}) e^{j\pi n} + X_3 H(e^{j3\pi/2}) e^{j3\pi/2n} \end{aligned}$$

$$y(n) = \frac{1}{4}H(e^{j0})e^{j0n} + \frac{1}{4}H(e^{j\pi/2})e^{j\pi/2n} + \frac{1}{4}H(e^{j\pi})e^{j\pi n} + \frac{1}{4}H(e^{j3\pi/2})e^{j3\pi/2n} \quad (4.45)$$

Given that

$$\begin{aligned} y(n) &= \cos\left(\frac{5\pi}{2}n + \frac{\pi}{4}\right) \\ &= \cos\left(2\pi n + \frac{\pi}{2}n + \frac{\pi}{4}\right) \\ &= \cos\left(\frac{\pi}{2}n + \frac{\pi}{4}\right) \end{aligned}$$

$$y(n) = \frac{1}{2}e^{j(\pi/2n + \pi/4)} + \frac{1}{2}e^{-j(\pi/2n + \pi/4)}$$

$$y(n) = \frac{1}{2}e^{j(\pi/2n + \pi/4)} + \frac{1}{2}e^{j(3\pi/2n - \pi/4)}$$

$$y(n) = \frac{1}{2}e^{j\pi/4}e^{j(\pi/2n)} + \frac{1}{2}e^{-j\pi/4}e^{j(3\pi/2n)}$$

Comparing the above equation with Eq. (4.45), we get

$$H(e^{j0}) = H(e^{j\pi}) = 0,$$

$$H(e^{j\pi/2}) = 2e^{j\pi/4}, \quad H(e^{j3\pi/2}) = 2e^{-j\pi/4}$$

**Example 4.12** Determine the output of the filter shown in Fig. 4.9 for the following periodic inputs:

(a)  $x_1(n) = (-1)^n$

(b)  $x_2(n) = 1 + \sin\left(\frac{3\pi}{8}n + \frac{\pi}{4}\right)$

(c)  $x_3(n) = \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^{n-4k} u(n-4k)$

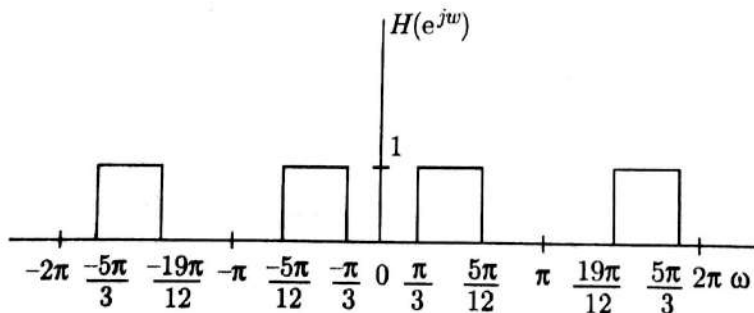


Fig. 4.9

### Solution

(a) Consider the given signal  $x_1(n)$ .

$$x_1(n) = (-1)^n = e^{j\pi n} = e^{j2\pi/2n}$$

Therefore,  $x_1(n)$  is periodic with period  $N = 2$ , and frequency  $\omega_0 = \frac{2\pi}{N} = \frac{2\pi}{2} = \pi$ . From Eq. (4.16), we have

$$\begin{aligned} x_1(n) &= \sum_{k=\langle N \rangle} X_k e^{jk\omega_0 n} \\ &= \sum_{k=0}^{N-1} X_k e^{jk\omega_0 n} = \sum_{k=0}^1 X_k e^{jk2\pi/2n} \\ x_1(n) &= X_0 + X_1 e^{j2\pi/2n} \end{aligned}$$

Comparing with

$$x_1(n) = e^{j2\pi/2n}$$

we get

$$X_0 = 0, \quad X_1 = 1$$

From Eq. (4.44), the output  $y_1(n)$  of the system is given by

$$\begin{aligned} y_1(n) &= \sum_{k=\langle N \rangle} X_k H(e^{jk\omega_0}) e^{jk\omega_0 n} \\ &= \sum_{k=0}^{N-1} X_k H(e^{jk2\pi/2}) e^{jk2\pi/2n} \\ &= \sum_{k=0}^1 X_k H(e^{jk2\pi/2}) e^{jk2\pi/2n} \\ &= 0 + X_1 H(e^{j\pi}) e^{j\pi n} \\ &= 0 \quad [\text{from Fig. 4.9, } H(e^{j\pi}) = 0] \end{aligned}$$

(b) Consider the given signal  $x_2(n)$

$$\begin{aligned} x_2(n) &= 1 + \sin\left(\frac{3\pi}{8}n + \frac{\pi}{4}\right) \\ &= 1 + \sin(\omega n + \phi) \end{aligned}$$

The frequency  $\omega$  of the given signal is

$$\omega = \frac{3\pi}{8}$$

$$\frac{2\pi}{\omega} = \frac{16}{3} = \frac{N}{m} \quad \text{a rational number}$$

Therefore,  $x_2(n)$  is periodic with period  $N = 16$  and frequency  $\omega_0 = 2\pi/N = 2\pi/16$ . The signal  $x_2(n)$

may be written as

$$\begin{aligned}
 x_2(n) &= 1 + \frac{e^{j(3\pi/8n+\pi/4)} - e^{-j(3\pi/8n+\pi/4)}}{2j} \\
 &= e^{j02\pi/16n} - \frac{j}{2} e^{j\pi/4} e^{j32\pi/16n} \\
 &\quad + \frac{j}{2} e^{-j\pi/4} e^{-j32\pi/16n} \\
 &= X_0 e^{j02\pi/16n} + X_3 e^{j32\pi/16n} \\
 &\quad + X_{-3} e^{-j32\pi/16n}
 \end{aligned}$$

Noting that the DTFS coefficients are periodic with period  $N = 16$ , we have

$$\begin{aligned}
 X_k &= X_{k+N} \\
 X_{-3} &= X_{13}
 \end{aligned}$$

Therefore,

$$x_2(n) = X_0 e^{j02\pi/16n} + X_3 e^{j32\pi/16n} + X_{13} e^{j132\pi/16n}$$

The nonzero Fourier coefficients of  $x_2(n)$  in the range  $0 \leq k \leq 15$  are

$$X_0 = 1, \quad X_3 = -\frac{j}{2} e^{j\pi/4}, \quad X_{13} = \frac{j}{2} e^{-j\pi/4}$$

From Eq. (4.44), the output  $y_2(n)$  of the system is given by

$$\begin{aligned}
 y_2(n) &= \sum_{k=\langle N \rangle} X_k H(e^{jk\omega_0}) e^{jk\omega_0 n} \\
 &= \sum_{k=0}^{N-1} X_k H(e^{jk2\pi/16}) e^{jk2\pi/16n} \\
 &= \sum_{k=0}^{15} X_k H(e^{jk2\pi/16}) e^{jk2\pi/16n} \\
 &= X_0 H(e^{j02\pi/16}) e^{j02\pi/16n} \\
 &\quad + X_3 H(e^{j32\pi/16}) e^{j32\pi/16n} \\
 &\quad + X_{13} H(e^{j132\pi/16}) e^{j132\pi/16n} \\
 &= X_0 H(e^{j0\pi/8}) e^{j02\pi/16n} + X_3 H(e^{j3\pi/8}) e^{j32\pi/16n} \\
 &\quad + X_{13} H(e^{j13\pi/8}) e^{j132\pi/16n}
 \end{aligned}$$

from Fig. 4.9, we have

$$H(e^{j0\pi/8}) = 0 \quad \text{and} \quad H(e^{j3\pi/8}) = H(e^{j13\pi/8}) = 1$$

Therefore,

$$\begin{aligned}
 y_2(n) &= 0 - \frac{j}{2} e^{j\pi/4} e^{j32\pi/16n} + \frac{j}{2} e^{-j\pi/4} e^{-j32\pi/16n} \\
 &= \frac{e^{j(3\pi/8n+\pi/4)} - e^{-j(3\pi/8n+\pi/4)}}{2j}
 \end{aligned}$$

$$y_2(n) = \sin(3\pi/8n + \pi/4)$$

(c) The given signal  $x_3(n)$  may be written as

$$\begin{aligned}
 x_3(n) &= \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^{n-4k} u(n-4k) \\
 x_3(n) &= \left(\frac{1}{2}\right)^n u(n) * \left(\sum_{k=-\infty}^{\infty} \delta(n-4k)\right) \\
 x_3(n) &= g_1(n) * g_2(n)
 \end{aligned}$$

where

$$\begin{aligned}
 g_1(n) &= \left(\frac{1}{2}\right)^n u(n) \\
 g_2(n) &= \left(\sum_{k=-\infty}^{\infty} \delta(n-4k)\right)
 \end{aligned}$$

Let  $h(n)$  be the impulse response of the given filter. The output  $y_3(n)$  of the filter is

$$\begin{aligned}
 y_3(n) &= x_3(n) * h(n) \\
 &= [g_1(n) * g_2(n)] * h(n) \\
 y_3(n) &= g_1(n) * [g_2(n) * h(n)] = g_1(n) * z(n)
 \end{aligned}$$

Therefore,  $y_3(n)$  may be obtained by passing the signal  $g_2(n)$  through the filter with impulse response  $h(n)$  (frequency response  $H(e^{j\omega})$ ), and then convolving the result with  $g_1(n)$ .

The signal  $g_2(n)$  is periodic with period  $N = 4$  and frequency  $\omega_0 = 2\pi/N = 2\pi/4$ . The given signal for one period may be written as

$$\begin{aligned}
 g_2(n) &= \begin{cases} 1, & n = 0 \\ 0, & n = 1, 2, 3 \end{cases} \\
 &= \delta(n)
 \end{aligned}$$

Therefore, the Fourier series coefficients are

$$\begin{aligned} X_k &= \frac{1}{N} \sum_{n=0}^{N-1} g_2(n) e^{-jk\omega_0 n} \\ &= \frac{1}{4} \sum_{n=0}^3 g_2(n) e^{-jk2\pi/4n} \\ &= \frac{1}{4} \sum_{n=0}^3 \delta(n) e^{-jk2\pi/4n} \\ X_k &= \frac{1}{4} \quad \text{for all } k \end{aligned}$$

The output  $z(n)$  obtained by passing the periodic signal  $g_2(n)$  through the filter with frequency response  $H(e^{j\omega})$  is given by Eq. (4.44).

$$\begin{aligned} z(n) &= \sum_{k=\langle N \rangle} X_k H(e^{jk\omega_0}) e^{jk\omega_0 n} \\ &= \sum_{k=0}^{N-1} X_k H(e^{jk2\pi/N}) e^{jn k 2\pi/N} \\ &= \sum_{k=0}^3 X_k H(e^{jk2\pi/4}) e^{jn k 2\pi/4} \\ z(n) &= X_0 H(e^{j0}) e^{j0} + X_1 H(e^{j\pi/2}) e^{jn\pi/2} \\ &\quad + X_2 H(e^{j\pi}) e^{jn\pi} + X_3 H(e^{j3\pi/2}) e^{jn3\pi/2} \\ &= \frac{1}{4} \left[ H(e^{j0}) e^{j0} + H(e^{j\pi/2}) e^{jn\pi/2} \right. \\ &\quad \left. + H(e^{j\pi}) e^{jn\pi} + H(e^{j3\pi/2}) e^{jn3\pi/2} \right] \end{aligned}$$

From Fig. 4.9, we have

$$H(e^{j0}) = H(e^{j\frac{\pi}{2}}) = H(e^{j\pi}) = H(e^{j\frac{3\pi}{2}}) = 0$$

Therefore,

$$z(n) = 0$$

Now, the final output is

$$\begin{aligned} y_3(n) &= g_1(n) * z(n) \\ &= g_1(n) * 0 = 0 \end{aligned}$$

**Example 4.13** In each of the following, we specify the DTFS coefficients of a signal that is periodic with period  $N = 8$ . Determine the signal  $x(n)$  in each case.

$$(a) X_k = \cos\left(\frac{k\pi}{4}\right) + \sin\left(\frac{3k\pi}{4}\right)$$

$$(b) X_k = \begin{cases} \sin\left(\frac{k\pi}{3}\right), & 0 \leq k \leq 6 \\ 0, & k = 7 \end{cases}$$

**Solution**

(a) The signal  $x(n)$  is periodic with period  $N = 8$  and frequency  $\omega_0 = 2\pi/N = 2\pi/8$ . The DTFS coefficients for one period  $0 \leq k \leq 7$  are

$$X_k = \cos\left(\frac{k\pi}{4}\right) + \sin\left(\frac{3k\pi}{4}\right)$$

$$X_0 = 1, \quad X_1 = \sqrt{2}, \quad X_2 = -1, \quad X_3 = 0$$

$$X_4 = -1, \quad X_5 = -\sqrt{2}, \quad X_6 = 1, \quad X_7 = 0$$

From the synthesis Eq. (4.16), over one period  $0 \leq n \leq 7$ , we have

$$\begin{aligned} x(n) &= \sum_{k=\langle N \rangle} X_k e^{jk\omega_0 n} \\ &= \sum_{k=0}^7 X_k e^{jk\omega_0 n} \\ &= X_0 + X_1 e^{j\omega_0 n} + X_2 e^{j2\omega_0 n} + X_3 e^{j3\omega_0 n} \\ &\quad + X_4 e^{j4\omega_0 n} + X_5 e^{j5\omega_0 n} + X_6 e^{j6\omega_0 n} \\ &\quad + X_7 e^{j7\omega_0 n} \\ &= 1 + \sqrt{2} e^{j\omega_0 n} - e^{j2\omega_0 n} - e^{j4\omega_0 n} \\ &\quad - \sqrt{2} e^{j5\omega_0 n} + e^{j6\omega_0 n} \\ x(n) &= 1 + \sqrt{2} e^{j2\pi/8n} - e^{j22\pi/8n} - e^{j42\pi/8n} \\ &\quad - \sqrt{2} e^{j52\pi/8n} + e^{j62\pi/8n} \end{aligned}$$

$$x(n) = \begin{cases} 4, & n = 1, 7 \\ 4j, & n = 3 \\ -4j, & n = 5 \\ 0, & n = 0, 2, 4, 6 \end{cases}$$

Over one period  $0 \leq n \leq 7$ ,  $x(n)$  may be written as

$$\begin{aligned} x(n) &= 4\delta(n-1) + 4j\delta(n-3) - 4j\delta(n-5) \\ &\quad + 4\delta(n-7) \end{aligned}$$

(b) The signal  $x(n)$  is periodic with period  $N = 8$  and frequency  $\omega_0 = 2\pi/N = 2\pi/8$ . The DTFS coefficients for one period  $0 \leq k \leq 7$  are

$$X_k = \begin{cases} \sin\left(\frac{k\pi}{3}\right), & 0 \leq k \leq 6 \\ 0, & k = 7 \end{cases}$$

From the synthesis Eq. (4.16), over one period  $0 \leq n \leq 7$ , we have

$$\begin{aligned} x(n) &= \sum_{k=(N)} X_k e^{jk\omega_0 n} \\ &= \sum_{k=0}^7 X_k e^{jk\omega_0 n} \\ &= \sum_{k=0}^6 \sin\left(\frac{k\pi}{3}\right) e^{jk\omega_0 n} \end{aligned}$$

$$\begin{aligned} x(n) &= \sum_{k=0}^6 \frac{1}{2j} (e^{jk\pi/3} - e^{-jk\pi/3}) e^{jk2\pi/8n} \\ &= \frac{1}{2j} \sum_{k=0}^6 e^{jk\pi/3} e^{jk\pi/4n} - \frac{1}{2j} \sum_{k=0}^6 e^{-jk\pi/3} e^{jk\pi/4n} \\ &= \frac{1}{2j} \sum_{k=0}^6 e^{j(\pi/4n + \pi/3)k} - \frac{1}{2j} \sum_{k=0}^6 e^{j(\pi/4n - \pi/3)k} \\ &= \frac{1}{2j} \frac{1 - e^{j7(\pi/4n + \pi/3)}}{1 - e^{j(\pi/4n + \pi/3)}} - \frac{1}{2j} \frac{1 - e^{j7(\pi/4n - \pi/3)}}{1 - e^{j(\pi/4n - \pi/3)}} \end{aligned}$$

$$\begin{aligned} x(n) &= \frac{1}{2j} \left[ \frac{-e^{j3n\pi/4} \sin \frac{7}{2}(\pi/4n + \pi/3)}{\sin \frac{1}{2}(\pi/4n + \pi/3)} \right. \\ &\quad \left. + \frac{e^{j3n\pi/4} \sin \frac{7}{2}(\pi/4n - \pi/3)}{\sin \frac{1}{2}(\pi/4n - \pi/3)} \right] \end{aligned}$$

$$\begin{aligned} x(n) &= \frac{e^{j3n\pi/4}}{2j} \left[ \frac{\sin \frac{7}{2}(\pi/4n - \pi/3)}{\sin \frac{1}{2}(\pi/4n - \pi/3)} \right. \\ &\quad \left. - \frac{\sin \frac{7}{2}(\pi/4n + \pi/3)}{\sin \frac{1}{2}(\pi/4n + \pi/3)} \right] \end{aligned}$$

**Example 4.14** Consider the following three discrete-time signals with a fundamental period of  $N = 6$ :

$$x(n) = 1 + \cos\left(\frac{2\pi}{6}n\right),$$

$$y(n) = \sin\left(\frac{2\pi}{6}n + \frac{\pi}{4}\right),$$

$$z(n) = x(n)y(n)$$

(a) Determine the Fourier series coefficients of  $x(n)$ .

(b) Determine the Fourier series coefficients of  $y(n)$ .

(c) Use the results of parts (a) and (b), along with the multiplication property of the discrete-time Fourier series, to determine the Fourier coefficients of  $z(n) = x(n)y(n)$ .

### Solution

(a) The given signal  $x(n)$  is periodic with period  $N = 6$  and frequency  $\omega_0 = 2\pi/6$ . Consider the given signal  $x(n)$ ,

$$\begin{aligned} x(n) &= 1 + \cos(2\pi/6n) = 1 + \frac{1}{2}e^{j2\pi/6n} \\ &\quad + \frac{1}{2}e^{-j2\pi/6n} \end{aligned}$$

$$x(n) = X_0 + X_1 e^{j2\pi/6n} + X_{-1} e^{-j2\pi/6n}$$

Noting that the DTFS coefficients are periodic with period  $N = 6$ , we have

$$X_k = X_{k+N}$$

$$X_{-1} = X_5$$

Therefore, over the interval  $0 \leq k \leq 5$  the nonzero DTFS coefficients of  $x(n)$  are

$$X_0 = 1, \quad X_1 = \frac{1}{2}, \quad X_5 = \frac{1}{2}$$

(b) The given signal  $y(n)$  is periodic with period  $N = 6$  and frequency  $\omega_0 = 2\pi/6$ . Consider the given signal  $y(n)$ ,

$$\begin{aligned} y(n) &= \sin\left(\frac{2\pi}{6}n + \frac{\pi}{4}\right) \\ &= \frac{1}{2j} e^{j(2\pi/6n + \pi/4)} - \frac{1}{2j} e^{-j(2\pi/6n + \pi/4)} \\ &= \frac{1}{2j} e^{j\pi/4} e^{j2\pi/6n} - \frac{1}{2j} e^{-j\pi/4} e^{-j2\pi/6n} \\ &= Y_1 e^{j2\pi/6n} + Y_{-1} e^{-j2\pi/6n} \end{aligned}$$

Noting that the DTFS coefficients are periodic with period  $N = 6$ , we have

$$Y_k = Y_{k+N}$$

$$Y_{-1} = Y_5$$

Therefore, over the interval  $0 \leq k \leq 5$  the nonzero DTFS coefficients of  $y(n)$  are

$$Y_1 = \frac{1}{2j} e^{j\pi/4}, \quad Y_5 = -\frac{1}{2j} e^{-j\pi/4}$$

(c) The given signal  $z(n) = x(n)y(n)$  is periodic with period  $N = 6$ . Using the multiplication property Eq. (4.30), we have

$$z(n) = x(n)y(n)$$

$$Z_k = \sum_{r=\langle N \rangle} X_r Y_{k-r}$$

$$Z_k = \sum_{r=0}^{N-1} X_r Y_{k-r}$$

$$Z_k = \sum_{r=0}^5 X_r Y_{k-r}$$

$$Z_k = X_0 Y_k + X_1 Y_{k-1} + X_2 Y_{k-2} + X_3 Y_{k-3} + X_4 Y_{k-4} + X_5 Y_{k-5}$$

$$Z_k = X_0 Y_k + X_1 Y_{k-1} + X_5 Y_{k-5}$$

Making the use of  $Y_k = Y_{k+N}$ , we get the DTFS coefficients of  $z(n)$  over the interval  $0 \leq k \leq 5$ .

$$Z_k = \begin{cases} \frac{1}{2} \sin\left(\frac{\pi}{4}\right), & k = 0 \\ \frac{1}{2j} e^{j\frac{\pi}{4}}, & k = 1 \\ \frac{1}{4j} e^{j\frac{\pi}{4}}, & k = 2 \\ 0, & k = 3 \\ -\frac{1}{4j} e^{-j\frac{\pi}{4}}, & k = 4 \\ -\frac{1}{2j} e^{-j\frac{\pi}{4}}, & k = 5 \end{cases}$$

**Example 4.15** Suppose we are given the following information about a periodic signal  $x(n)$  with period  $N = 8$  and Fourier series coefficients  $X_k$ :

1.  $X_k = -X_{k-4}$
2.  $x(2n+1) = (-1)^n$

Sketch one period of  $x(n)$ .

### Solution

We know that if

$$x(n) \longleftrightarrow X_k$$

then, from the frequency shifting property, i.e., Eq. (4.23), we have

$$(-1)^n x(n) = e^{j\pi n} x(n) = e^{j2\pi/N N/2 n} x(n) \longleftrightarrow X_{k-N/2}$$

In this case  $N = 8$ , therefore

$$(-1)^n x(n) \longleftrightarrow X_{k-4}$$

Since, it is given that  $X_k = -X_{k-4}$ , we have

$$(-1)^n x(n) \longleftrightarrow -X_k$$

$$(-1)^n x(n) = -x(n)$$

$$x(n)[1 + (-1)^n] = 0$$

This implies that

$$x(n) = 0, \text{ for } n = 0, \pm 2, \pm 4, \pm 6, \dots$$

From the given information  $x(2n+1) = (-1)^n$ , we get

$$x(1) = 1, \quad x(3) = -1, \quad x(5) = 1, \quad x(7) = -1$$

Therefore, over one period  $0 \leq n \leq 7$ ,  $x(n)$  is defined as

$$x(n) = \begin{cases} 0, & n = 0, 2, 4, 6 \\ 1, & n = 1, 5 \\ -1, & n = 3, 7 \end{cases}$$

One period of  $x(n)$  is shown in Fig. 4.10.

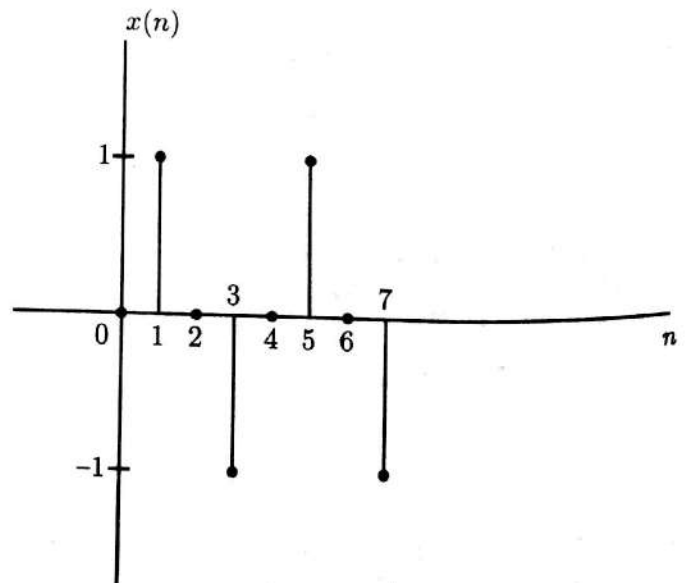


Fig. 4.10

**Example 4.16** Let  $x(n)$  be a periodic signal with period  $N = 8$  and Fourier series coefficients  $X_k = -X_{k-4}$ . A signal

$$y(n) = \left( \frac{1 + (-1)^n}{2} \right) x(n-1)$$

with period  $N = 8$  is generated. Denoting the Fourier series coefficients of  $y(n)$  by  $Y_k$ , find a function such that

$$Y_k = f(k)X_k$$

### Solution

We know that if

$$x(n) \longleftrightarrow X_k$$

then, from the frequency shifting property, i.e., Eq. (4.23), we have

$$(-1)^n x(n) = e^{j\pi n} x(n) = e^{j2\pi/N \cdot N/2n} \longleftrightarrow X_{k-N/2}$$

In this case  $N = 8$ , therefore

$$(-1)^n x(n) \longleftrightarrow X_{k-4}$$

Since, it is given that  $X_k = -X_{k-4}$ , we have

$$(-1)^n x(n) \longleftrightarrow -X_k$$

$$(-1)^n x(n) = -x(n)$$

$$-(-1)^n x(n) = x(n)$$

$$(-1)^{n+1} x(n) = x(n)$$

$$(-1)^n x(n-1) = x(n-1)$$

Now consider

$$y(n) = \left( \frac{1 + (-1)^n}{2} \right) x(n-1)$$

$$= \frac{1}{2} x(n-1) + \frac{1}{2} (-1)^n x(n-1)$$

We have already proved that  $(-1)^n x(n-1) = x(n-1)$ , therefore

$$y(n) = \frac{1}{2} x(n-1) + \frac{1}{2} x(n-1)$$

$$y(n) = x(n-1)$$

Using the time-shifting property, we get

$$Y_k = X_k e^{-jk2\pi/8} = f(k)X_k$$

Thus,

$$f(k) = e^{-jk2\pi/8}$$

## SUMMARY

1. A periodic discrete-time signal  $x(n)$  with period  $N$  can be represented by the discrete-time Fourier series (DTFS)

$$x(n) = \sum_{k=\langle N \rangle} X_k e^{jk\omega_0 n}$$

where  $\omega_0 = 2\pi/N$  and  $\sum_{k=\langle N \rangle}$  means summation over any range of consecutive  $k$ 's exactly  $N$  in length.

2. The DTFS coefficients  $X_k$  are given by

$$X_k = \frac{1}{N} \sum_{k=\langle N \rangle} x(n) e^{-jk\omega_0 n}$$

3. The coefficients  $X_k$  are periodic with period  $N$ , so that

$$X_k = X_{k+N}$$

4. The DTFS is a finite sum over only  $N$  terms. It provides an exact alternative representation of the

time signal, and issues such as convergence or the Gibbs phenomenon do not arise.

5. If  $X_k$  are the DTFS coefficients of the signal  $x(n)$ , then the coefficients of  $x(n-m)$  are equal to  $X_k e^{-j\omega_0 km}$ .
6. If the periodic sequence  $x(n)$  with DTFS coefficients  $X_k$  is input into an LTI system with impulse response  $h(n)$ , the DTFS coefficients  $Y_k$  of the output  $y(n)$  are given by

$$Y_k = X_k H(e^{j\omega_0 k})$$

where

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n) e^{-j\omega n}$$

7. If an LTI system is excited by a periodic signal, the response is also a periodic signal with the same fundamental period.

# MULTIPLE-CHOICE QUESTIONS

- If the Fourier series coefficients of a signal are periodic then the signal must be
  - continuous-time, periodic
  - discrete-time, periodic
  - continuous-time, non-periodic
  - discrete-time, non-periodic
- The DTFS coefficients of a real and even periodic signal are
  - real and odd
  - imaginary and even
  - real and even
  - imaginary and odd
- The DTFS coefficients of a real and odd periodic signal are
  - real and odd
  - imaginary and even
  - real and even
  - imaginary and odd
- Let  $X_k$  represent the DTFS coefficients of the periodic sequence  $x(n)$  with period  $N$ . The DTFS coefficients of the signal  $(-1)^n x(n)$  in terms of  $X_k$  are
  - $X_k$
  - $X_{-k}$
  - $X_{k+\frac{N}{2}}$
  - $X_{k-\frac{N}{2}}$
- Let  $X_k$  represent the DTFS coefficients of the periodic sequence  $x(n)$  with period  $N$ . The DTFS coefficients of the signal  $x(n-1)$  in terms of  $X_k$  are
  - $X_k e^{-jk2\pi/N}$
  - $X_k e^{jk2\pi/N}$
  - $X_{k+1}$
  - $X_{k-1}$
- Let  $X_k$  represent the DTFS coefficients of the periodic sequence  $x(n)$  with period  $N$ . The DTFS coefficients of the signal  $x(n+1)$  in terms of  $X_k$  are
  - $X_k e^{-jk2\pi/N}$
  - $X_k e^{jk2\pi/N}$
  - $X_{k+1}$
  - $X_{k-1}$
- Let  $X_k$  represent the DTFS coefficients of the periodic sequence  $x(n)$  with period  $N$ . The DTFS coefficients of the signal  $x\left(\frac{n}{2}\right)$  in terms of  $X_k$  are
  - $X_{2k}$
  - $X_{k/2}$
  - $2X_k$
  - $\frac{X_k}{2}$

## PROBLEMS

4.1 One period of the DTFS coefficients of a signal is given by

$$X_k = \left(\frac{1}{2}\right)^k, \quad 0 \leq k \leq 9$$

Find the time-domain signal  $x(n)$  assuming  $N = 10$ .

4.2 Find the time-domain signal corresponding to the DTFS coefficients

$$X_k = \cos\left(\frac{k4\pi}{11}\right) + 2j \sin\left(\frac{k6\pi}{11}\right)$$

4.3 Determine the DTFS coefficients for the following signals:

(a)  $x(n) = 1 + \sin(n\pi/12 + 3\pi/8)$

(b)  $x(n) = \cos(n\pi/30) + 2\sin(n\pi/90)$

4.4 Find the DTFS coefficients of the signal depicted in Fig. 4.11.

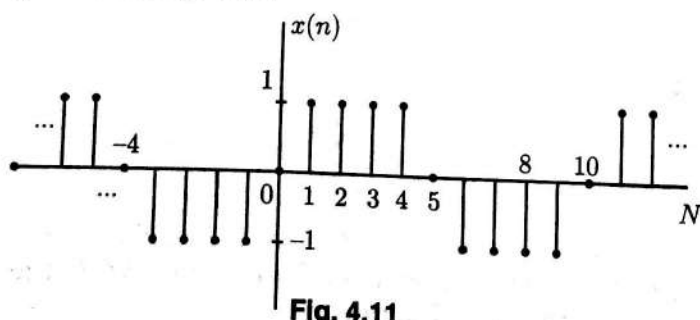


Fig. 4.11

4.5 Consider a discrete-time LTI system with impulse response

$$h(n) = \left(\frac{1}{2}\right)^{|n|}$$

Find the Fourier series representation of the output  $y(n)$  for each of the following inputs:

(a)  $x(n) = \sum_{k=-\infty}^{\infty} \delta(n-4k)$

(b)  $x(n)$  is periodic with period 6 and

$$x(n) = \begin{cases} 1, & n = 0, \pm 1 \\ 0, & n = \pm 2, \pm 3 \end{cases}$$

4.6 Consider a discrete-time LTI system with impulse response

$$h(n) = \begin{cases} 1, & 0 \leq n \leq 2 \\ -1, & -2 \leq n \leq -1 \\ 0, & \text{otherwise} \end{cases}$$

Given that the input to this system is

$$x(n) = \sum_{k=-\infty}^{\infty} \delta(n-4k)$$

determine the Fourier series coefficients of the output  $y(n)$ .

4.7 Consider a discrete-time LTI system whose frequency response is

$$H(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \pi/8 \\ 0, & \pi/8 < |\omega| < \pi \end{cases}$$

Show that if the input  $x(n)$  to this system has a period  $N = 3$ , the output  $y(n)$  has only one nonzero Fourier series coefficient per period.

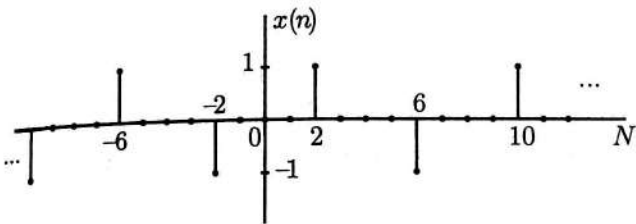


Fig. 4.12

4.8 Evaluate the DTFS coefficients of the following signals:

(a)  $x(n) = \cos\left(\frac{6\pi}{17}n + \frac{\pi}{3}\right)$

(b)  $x(n) = 2\sin\left(\frac{14\pi}{19}n\right) + \cos\left(\frac{10\pi}{19}n\right) + 1$

(c)  $x(n) = \sum_{m=-\infty}^{\infty} (-1)^m [\delta(n - 2m) + \delta(n + 3m)]$

(d)  $x(n)$  as depicted in Fig. 4.12.

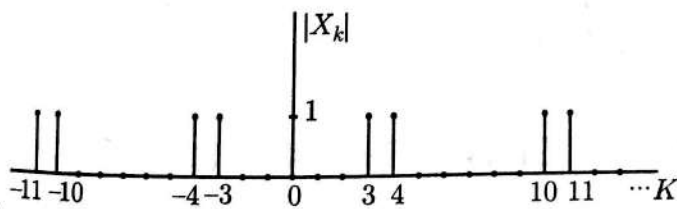


Fig. 4.13(a)

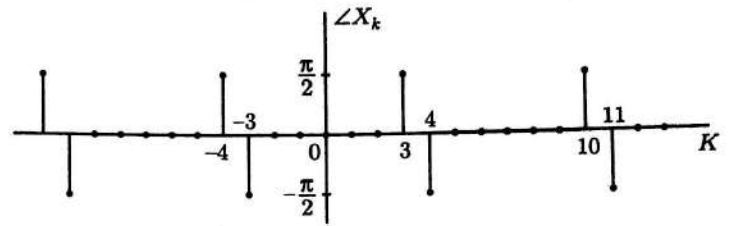


Fig. 4.13(b)

4.9 Evaluate the time-domain signals represented by the following DTFS coefficients:

(a)  $X_k = \cos\left(\frac{8\pi}{21}k\right)$

(b)  $X_k = \cos\left(\frac{10\pi}{19}k\right) + j2\sin\left(\frac{4\pi}{19}k\right)$

(c)  $X_k = \sum_{m=-\infty}^{\infty} (-1)^m [\delta(k - 2m) - 2\delta(k + 3m)]$

(d)  $X_k$  as depicted in Fig. 4.13.

4.10 One period of the DTFS coefficients of a signal is given by

$$X_k = \left(\frac{1}{2}\right)^k, \quad 0 \leq k \leq 9$$

Find the time-domain signal  $x(n)$  assuming  $N = 10$ .

4.11 Let  $X_k$  represent the DTFS coefficients of the periodic sequence  $x(n)$  with period  $N$ . Find the DTFS coefficients of each of the following signals in terms of  $X_k$ :

(a)  $x(n - n_0)$

(b)  $x(-n)$

(c)  $(-1)^n x(n)$

(d)  $y(n) = \left(\frac{1 + (-1)^n}{2}\right) x(n) = \begin{cases} x(n), & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$

(e)  $y(n) = \begin{cases} x(n), & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$

4.12 Show that for a real periodic sequence  $x(n)$ ,  $X_k = X_{N-k}^*$ .

## ANSWERS TO MULTIPLE-CHOICE QUESTIONS

1. (b)    2. (c)    3. (d)    4. (d)    5. (a)    6. (b)    7. (d)